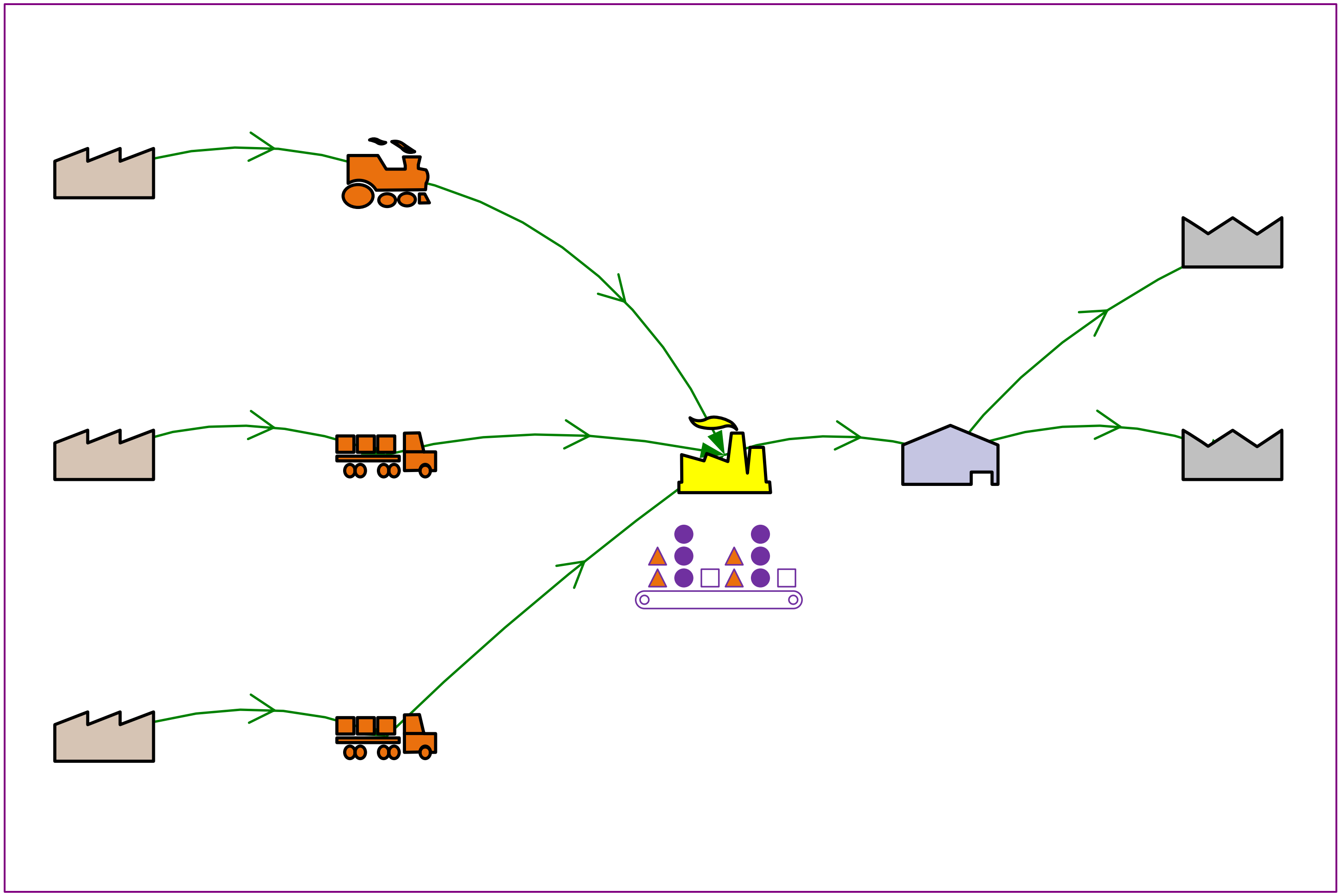
Mix Supply Network Mapping

Learn mix model supply network concepts and how to use eVSM Mix to build mix model maps for analyzing lead-times, cost, demand mix changes, comparing alternative suppliers, etc.



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How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

Table of Contents

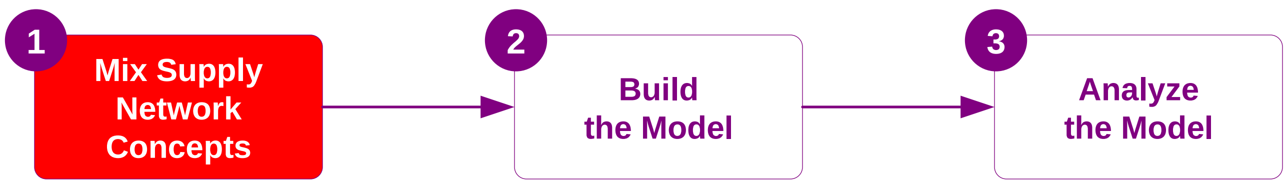
Mix Supply Network Concepts

The Mix Supply Network application provides logistics value stream mapping with calculation of demand, landed cost and lead times. It handles mapping of multiple products variants or those targeted for different regions.

Fast Draw and Mix Time are pre-requisites for this course. It would be a good idea to keep a printed copy of the Mix Time course handy as you work through this course.

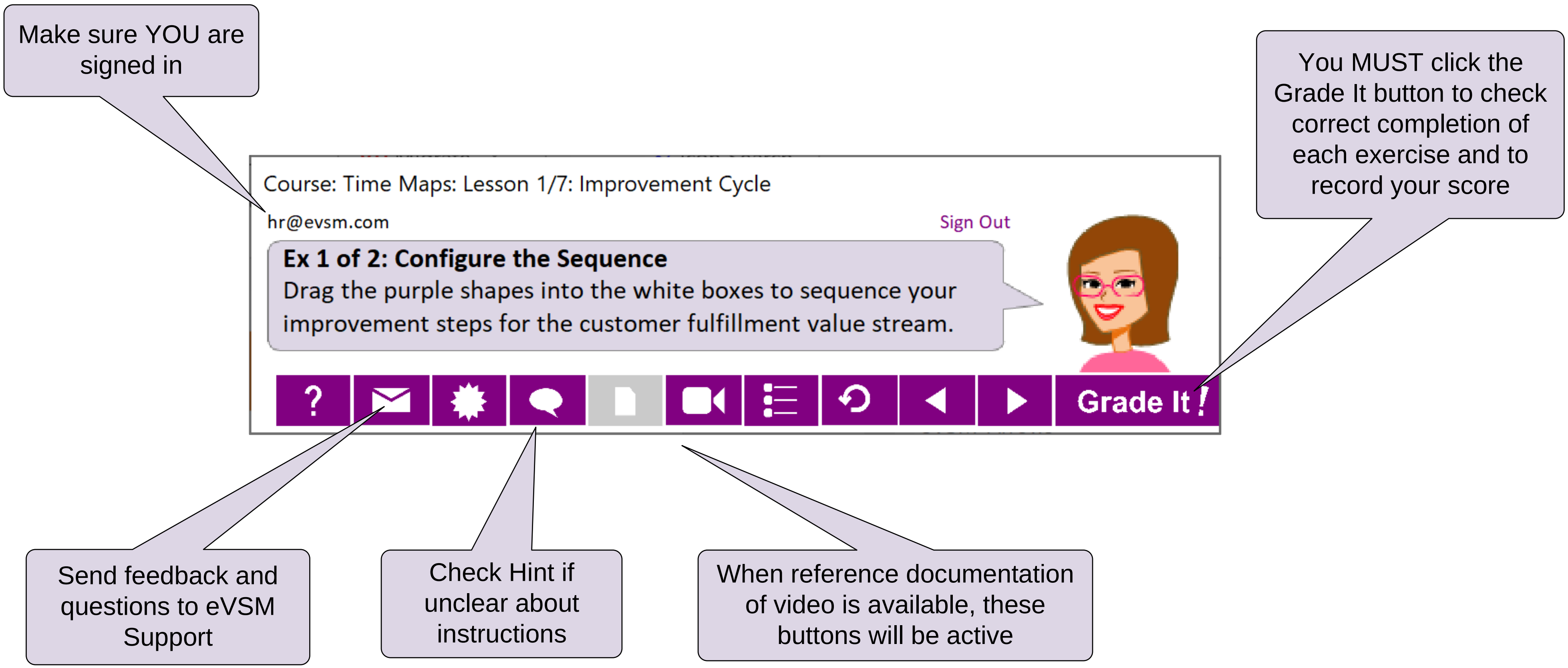
In this first lesson you will explore supply network mapping concepts and calculations.

Mix Supply Network – Learning Roadmap



Working with eLeanor

The eLeanor learning system includes a range of useful functions:



Important Notes

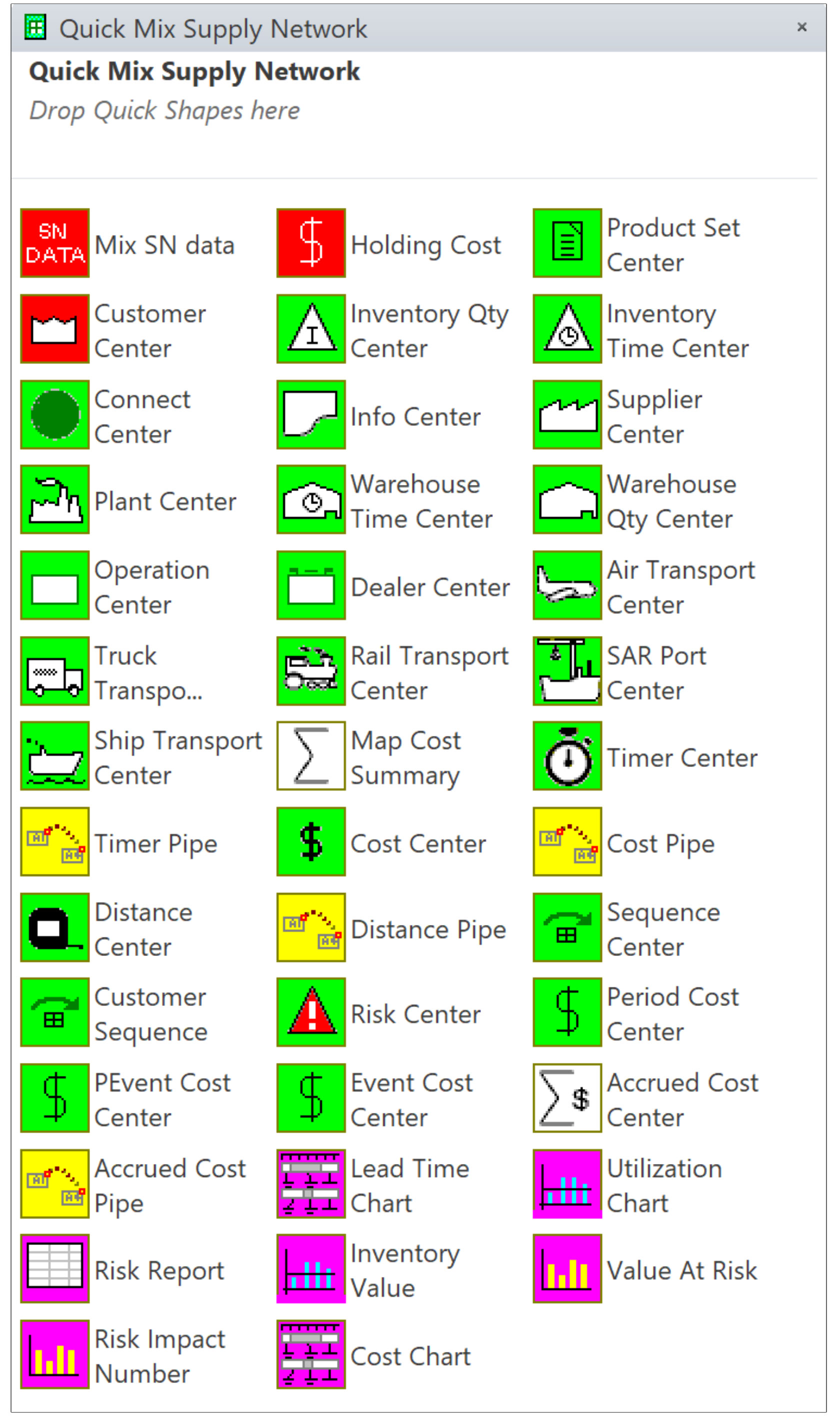
1. When you complete an exercise, you **MUST** click the “Grade It” button.
2. Points are deducted for incorrect attempts.
3. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button and ask your question.

eVSM Mix for Supply Network Design

Quick Mix Supply Network is a specialized supply network stencil designed to visualize and play "what-if" games to help improve the capacity, lead time, flexibility, and cost aspects of the network.

Benefits:

- Easily visualize, communicate, and discuss the supply network for a product family and geographic region.
- Input time/cost data for each leg of the network, and automatically generate lead time and cost charts.
- Do "what-if" studies with alternate routes for cost and time improvements or to reduce risk.
- Overlay maps onto geographic detail from any source and perhaps including mapping of adverse events like earthquakes or tornadoes.



Supply Network Example Map & Useful Features

Quantity

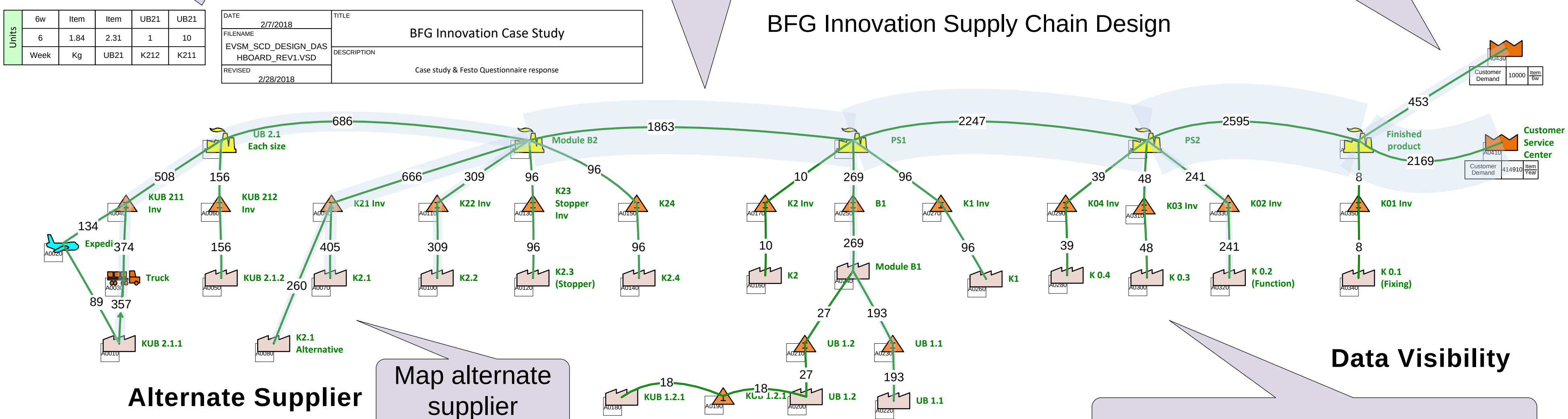
Understand supplier parts and assembly quantities

Cash River

See the costs per period from different sources building to the delivered cost

Demand

Map impact of demand of sub-components and final products on value stream



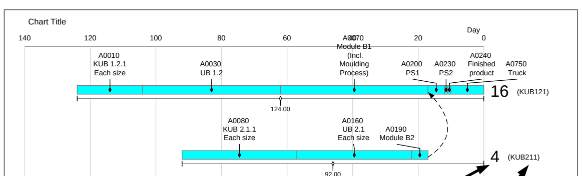
Alternate Supplier

Map alternate supplier strategies

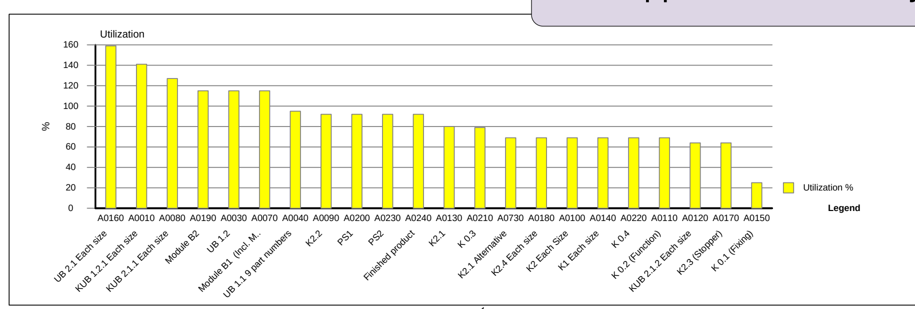
Data Visibility

Use hide/show to simplify the map but support the necessary analytics

Lead Times



Utilization



Lead time chart can be filtered by path numbers

Path Naming

Understand lead times and responsiveness

Lead Times

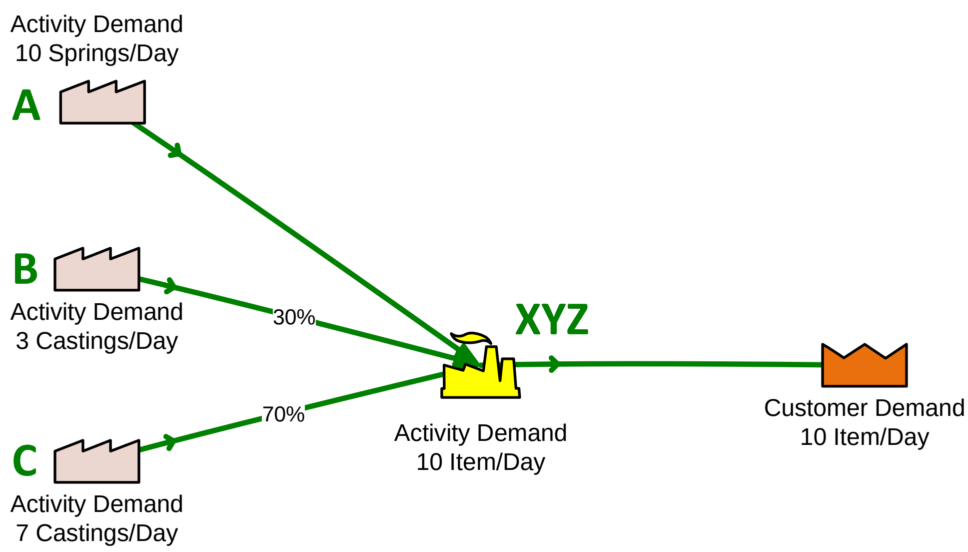
What-if studies to see what suppliers may become bottlenecks

Bottlenecks

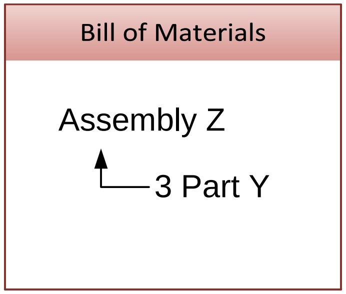
Upstream Demand Calculation

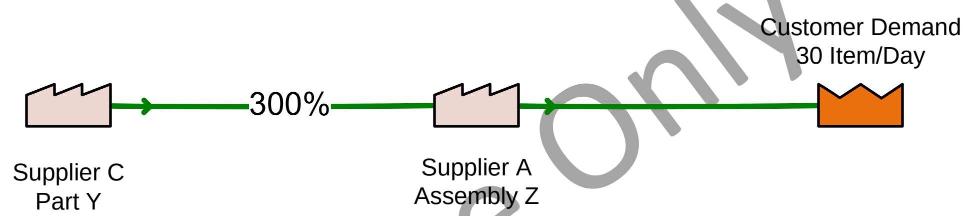
In a supply network, the total customer demand rolls upstream through to the suppliers.

- The demand is equal to the demand at the downstream node unless there is a demand % (other than 100) for the path. In this case the demand is factored by that percentage
- In the example supplier A has the same demand as Plant XYZ
- Supplier B’s demand is specified as 30% of the demand at plant XYZ



Upstream Demand Calculation





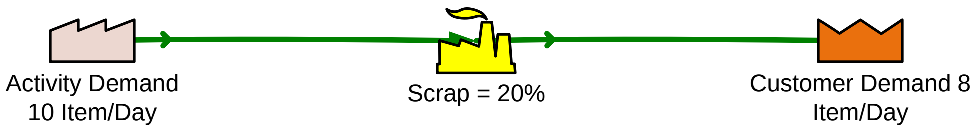
Each assembly Z needs 3 part Y's. This is noted with the 300% demand increase on the supply arrow

Q. What is the demand for Part Y at Supplier C

- 3 Item/Day
- 30 Item/Day
- 120 Item/Day
- 90 Item/Day
- 60 Item/Day

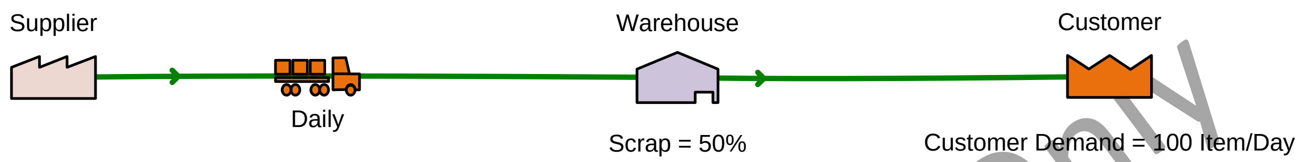
Scrap and Demand

Scrap at any point increases demand on the upstream nodes. So a 20% scrap changes a demand of 8 at the customer to a demand of 10 at the supplier:



Total Parts = Good Parts / (1-Scrap %)

Scrap and Demand



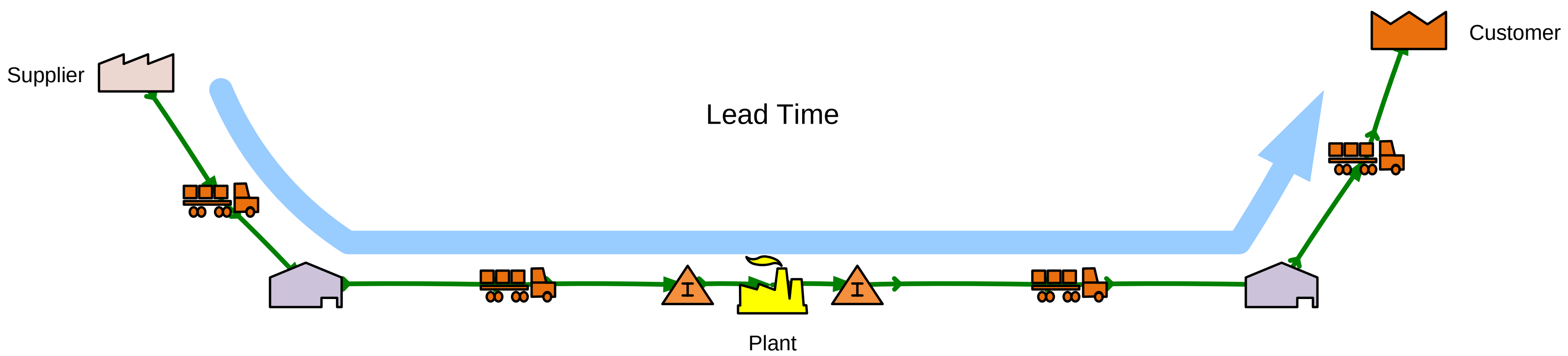
50% of supplier parts are considered defective when they are inspected at the warehouse.

Q. How many parts must the supplier ship to meet the customer demand of 100 good parts per day?

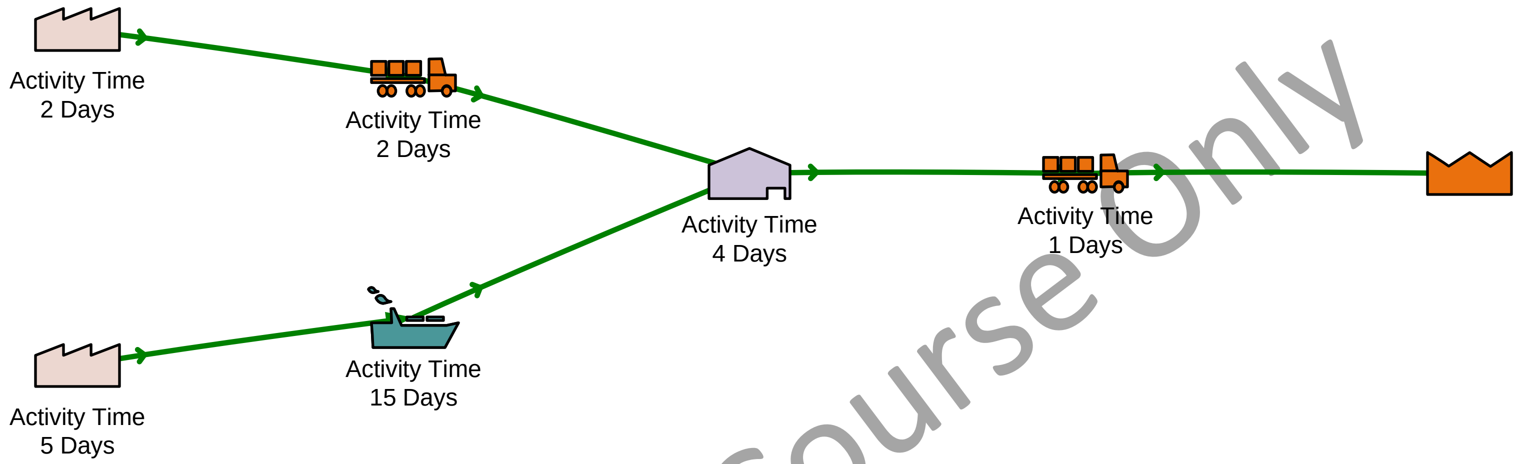
- 100
- 200
- 90
- 400
- 50

Lead Time

Lead time is the sum of the Activity Times through a segment of the network. This includes conversion times, wait times in inventory, and transport times:



Lead Time



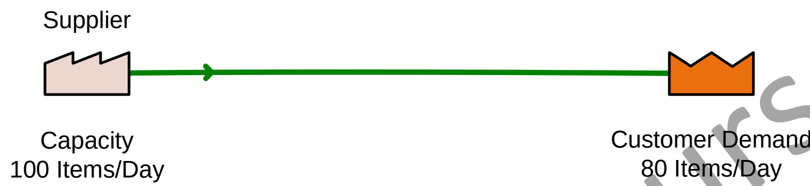
Q. If parts are "made to order" then what is the longest lead time from supplier to customer ?

- 29 Days
- 25 Days
- 34 Days
- 9 Days
- 5 Days

Capacity

The capacity of a facility is the maximum number of "good" products it can deliver in a given time period.

Capacity Utilization is the percentage of the maximum capacity that is currently in demand.



Q. A supplier is contracted to maintain a capacity of 100 Items/Day. The current demand is 80 Items/Day. What is the supplier utilization?

- 80%
- 100%
- 120%
- 55%
- 0%

Activity Time and Inventory Qty

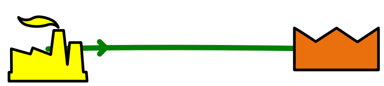
If an item takes, say 2 weeks to pass through a location, and the demand is 150 items/week, the minimum inventory at the location must be 2*150=300 items. So, the calculation for minimum inventory at a location is:

Minimum Inventory = Activity Time x Demand

Some nodes allow direct specification of inventory quantity. If the specified value is less then the calculated Minimum Inventory from the equation above, then the calculated value is used.

Look at the calculated “Inventory Count Used” at the bottom of the following examples via this logic.

Example 1

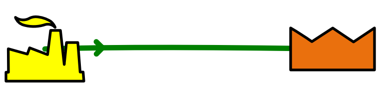


Activity Time = 2 Days
Inventory Qty = 18 Items

Demand
15 Item/Day

Inventory count used = 30 Items

Example 2



Activity Time = 2 Days
Inventory Qty = 47 Items

Demand
15 Item/Day

Inventory count used = 47 Items

Activity Time and Inventory

Q. What is the minimum inventory in the supply network shown, excluding any inventory at customer site?

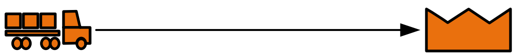
- 500 Items
- 200 Items
- 700 Items
- 50 Items
- 800 Items

Transport Inventory

The calculation of the average inventory in transit is simply

Average Inventory = Transport Time * Demand

This formula takes into account that when the transport vehicle is en route then its carrying enough inventory to sustain demand until the next time it is en route. Note that this average inventory calculation is INDEPENDENT of transport frequency

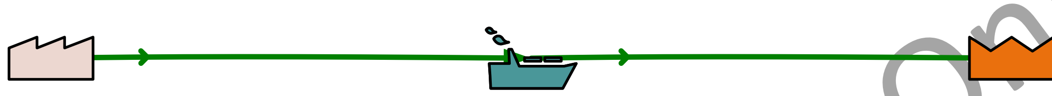


Transport Time = 2 Days **Demand = 100 Items/Day**

Average Inventory in Transit = Transport Time * Demand

= 2 * 100 = 200 Items

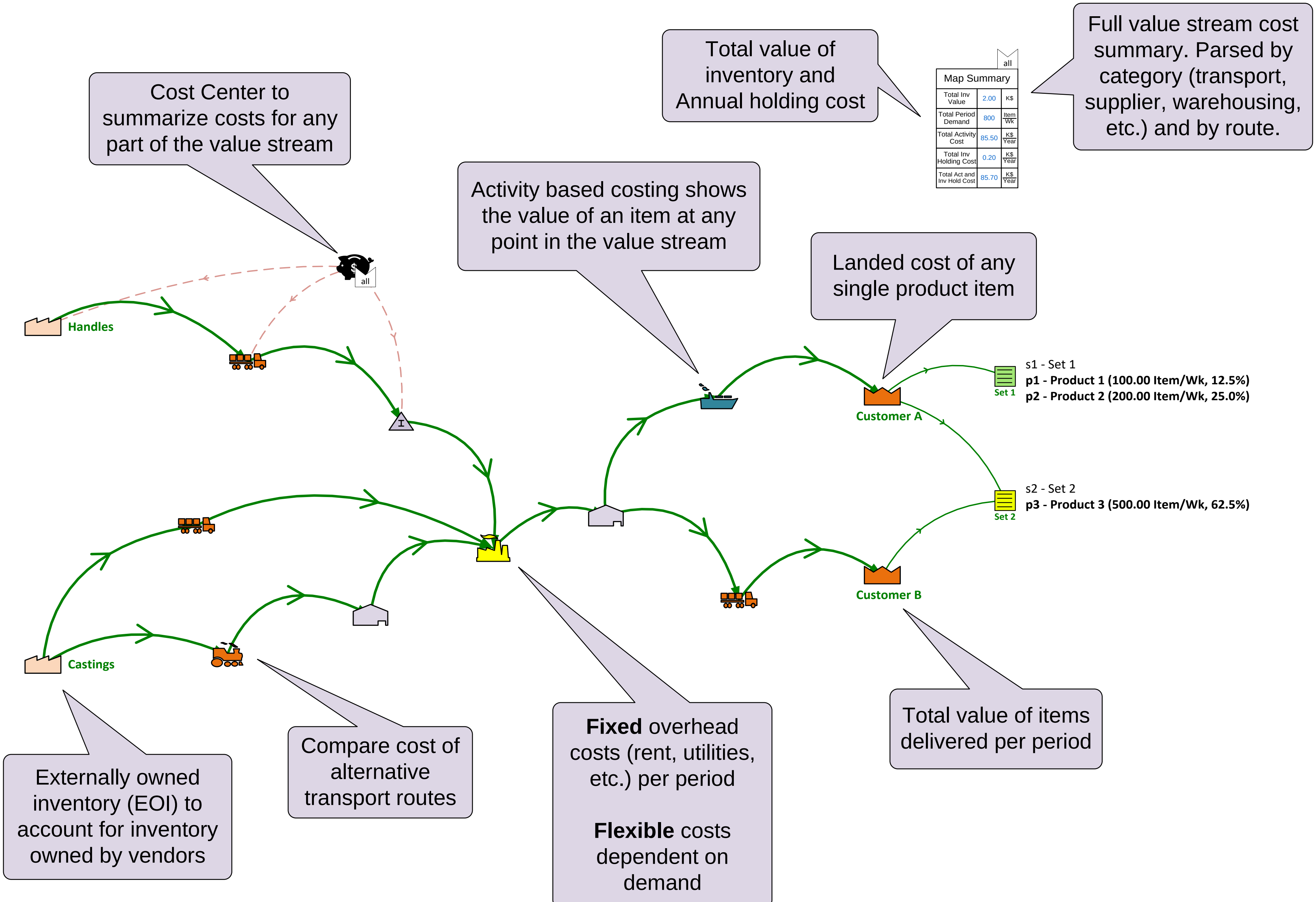
Transport Inventory



Q. On average how many items of inventory are in transit if the journey takes 15 days and ships arrive with frequency of once every 30 days

- 1500 Items
- 1200 Items
- 500 Items
- 3000 Items

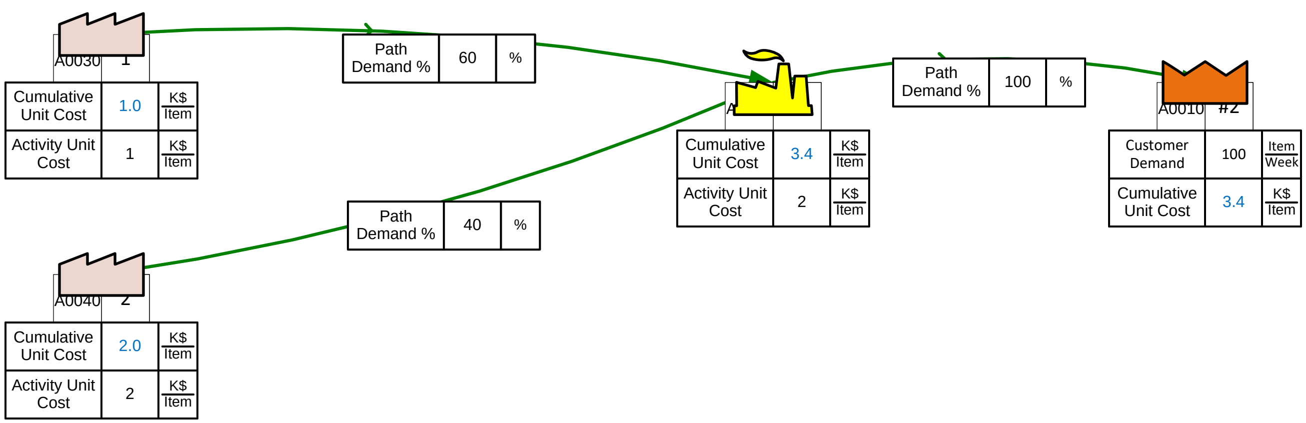
Cost concepts in Mix Supply Network



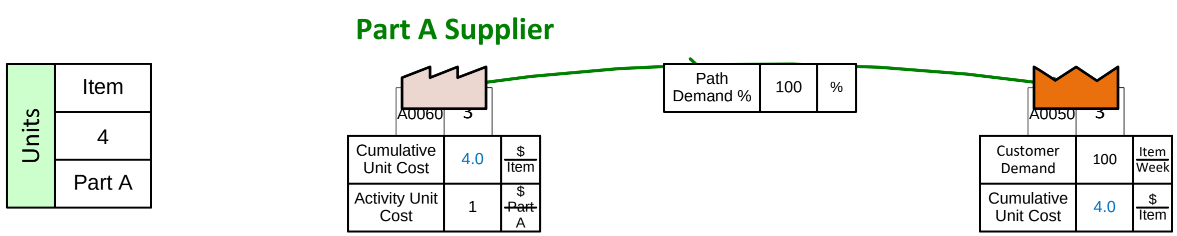
Unit Cost

In general, when two items arrive from different locations in the supply network to a node, their unit costs (part cost per unit of finished goods) are added together to calculate the unit cost at the node. This is the right behavior when the items are components that are both needed to make the product. There are exceptions

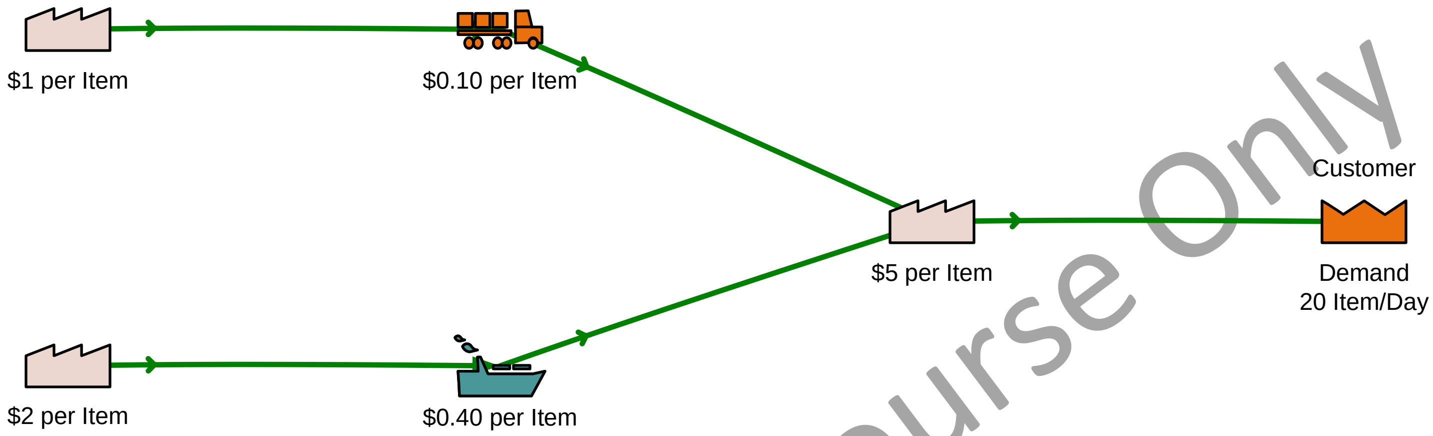
1. Sometimes the incoming items are the same but from alternate supply points. In this case demand percent values are specified for the path from each supply point. This is illustrated in the "cumulative unit cost" values for the example below. Note, the cumulative unit cost at the plant is: [[1 * 60] + [2 * 40]]/100. Add 2K$ to this for the plant activity and the total comes out to $3.4K$/Item.



2. When one finished good item needs multiple parts (e.g.. 6 screws), use a Units Converter to show how many parts go into the final product. In the example below a finished goods "Item" requires 4 of "Part A". Note how the price of 1 $/Part A equates to a unit cost of 4 $/Item at the customer.



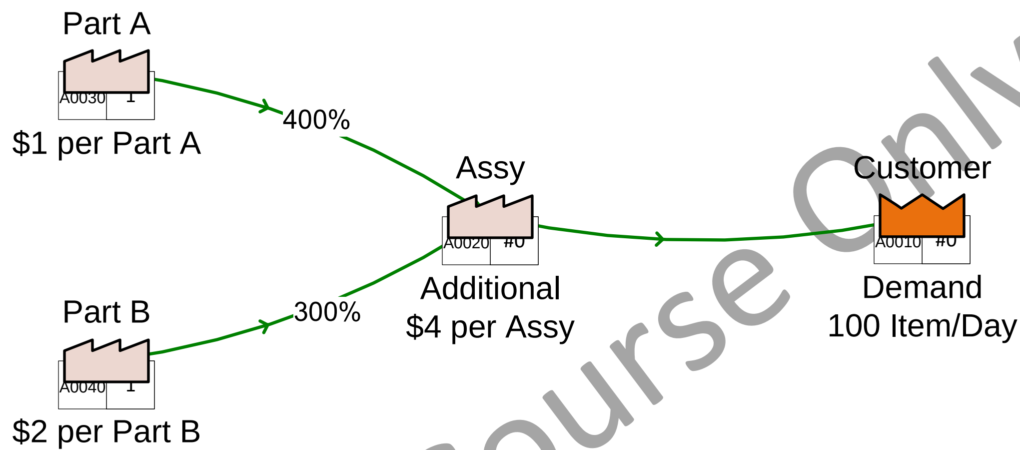
Unit Cost Exercise



Q. What is the approximate cost per item as delivered to the customer?

- $20.65 per Item
- $169 per Item
- $8.50 per Item
- $6.05 per Item

Total Cost per Day



Q. Each Assy needs Qty 4 of Part A and Qty 3 of Part B. Ignoring any shipping costs, what is the daily cost of parts delivered to the customer?

- ◎ $700 per Day
- ◎ $1700 per Day
- ◎ $1400 per Day
- ◎ $600 per Day

	Item	Item
Units	4	3
	Part A	Part B

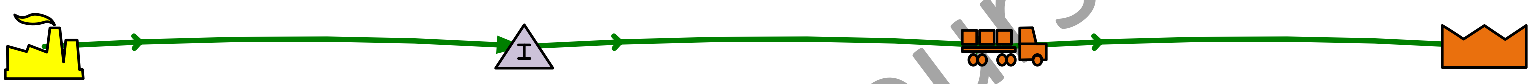
Holding Cost

If inventory of value is stored at a location, it has an associated “holding cost rate” per annum associated with financing its value and also related costs due to space, damage, theft etc.

Annual Holding Cost = Value of Inventory X Annual Holding Cost Rate

Example

Assume holding cost rate is 10%/year. In the below example, calculate the annual holding cost of inventory, excluding any inventory at the customer. Remember from exercise 5 that Activity Time is used to calculate minimum inventory.



Activity Time = 5 Day
Inventory Qty = 700 Item
Inventory Unit Value = $10/item

Inventory Qty = 200 Item
Inventory Unit Value = $10/Item

Activity Time = 5 Day
Inventory Unit Value = $20/item
(so Average transit inventory = 5*100=500 items)

Customer Demand = 100 Item/Day

Q. What is the annual holding cost of inventory?

- $1900
- $2100
- $1400
- $800

Externally Managed Inventory (EOI)

EOI means inventory financially owned by suppliers. This means that the supplier bears any related holding costs (finance, damage, theft..) for the inventory at that point

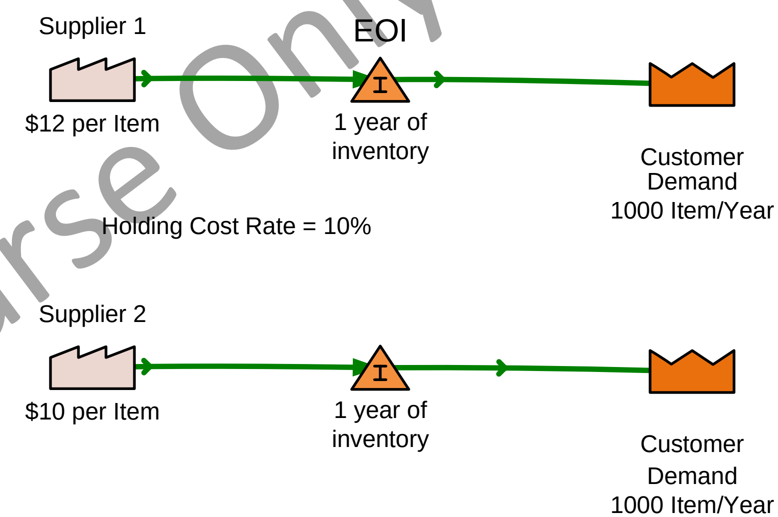
Example

The Customer is considering sourcing a part from two alternate suppliers, and requires 1 year worth of inventory to be maintained on-site.

Supplier 1 is willing to own all the inventory at the customer site until it is used, and has a delivered cost of $12/item.

Supplier 2 requires the inventory to be purchased and owned by the customer at a cost of $10/item.

The inventory annual holding cost rate is 10%/year. This means that if a location is storing $1000 of inventory value, it has an associated annual holding cost of $1000 * 0.1 = $100/year



Q. From a cost perspective, and with a annual holding cost rate of 10%, which would be the better solution, financially for the customer?

- ◎ Supplier 1
- ◎ Supplier 2
- ◎ Both the same

Risk in Supply Network and the Value at Risk calculation(VaR)

There can be risk associated with nodes on the network and a simple way to quantify each risk is with “Value at Risk” . Lets illustrate with a simple example

Suppose

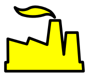
1. There is risk associated with a natural disaster

2. If the disaster occurs there is a one time loss of $2 million

3. The probability of the disaster occurring is 10%

Then the “Value at Risk” is calculated As $2 million * 10 percent = $200,000

Value at Risk (VaR)



Plant

Risk: Strike-based plant shutdown for a month

Probability of strike = 10%

Estimated impact of strike = $2 million

Q. What is the VaR (Value at Risk)?

- $2 million
- $20 million
- $0.2 million
- $0.8 million

Risk Impact Number (RIN)

Its useful to have a single number so risks can be ranked in an approximate way.

RIN (Risk Impact Number) = [Risk Probability] x [Risk Severity] x [Risk Detectability]

Each factor is on a scale of 1 to 10, where 10 represents highest probability, highest severity, or **lowest** detectability

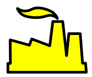
Example

Suppose:

1. The risk probability is 2
2. The risk severity is 7
3. The risk detectability is 6

Then the Risk Impact Number = 2 x 7 x 6 = 84

Risk Impact Number (RIN)



Plant

Risk: Strike-based plant shutdown for a month

Risk probability = 1 out of 10

Risk severity = 8 out of 10

Risk detectability = 2 out of 10

Q. What is the RIN?

- 1.6
- 16
- 160
- 1600

Risk Mitigation

A mitigating action if implemented can limit the risk and lower its probability of occurring but at a mitigating cost

Suppose:

- The original risk event has a cost of $2 million (if it happens) and a probability of occurrence of 10%
- There is a mitigating action that would reduce the probability of occurrence to 5% but would cost $50K to implement. The event if it still occurred despite mitigation would still cost $2 million
-

Then the original "Value of Risk" = $2million * 10% = $200,000

The mitigated "Value of Risk" = $2million * 5% = $100,000

Then the 'mitigated' reduction in "Value of Risk" would be calculated as $200K-$100K = $100K

This number would be considered in addition to the mitigation cost of $50K to help decide if mitigation action should be taken

Risk Mitigation

Plant

Risk: 10% probability of water shortage based plant shutdown in dry season (Impact of shortage is $2 million if it happens)

Mitigation Strategy: Reduce risk of water shortage through on-site holding tank

Cost of implementing holding tank: $30K

Mitigation Result: Risk Probability reduced to 5% (risk was originally 10%)

Q. What is the remaining "Value of Risk" after mitigation action is implemented ?

- $30K
- $100K
- $130K
- $70K

POE – Perfect Order Execution

A perfect order from a supplier is one that contains the right quantity being delivered of the right quality at the right time

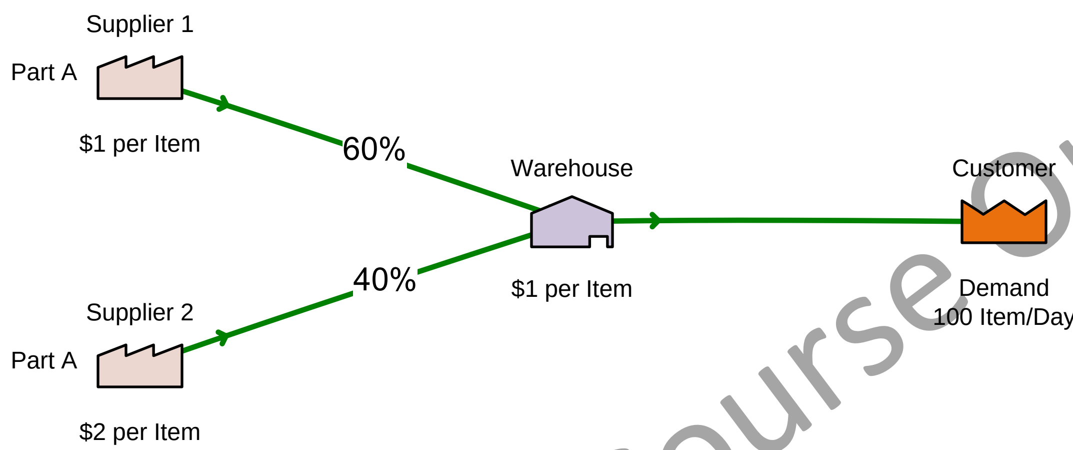
POE = [Right Quantity] x [Right Quality] x [Right Time]

Q. What is the approximate POE% for this supplier?

- Right Quantity: 99%
- Right Quality: 95%
- Right Time: 100%

- 5%
- 99%
- 92%
- 97%
- 94%

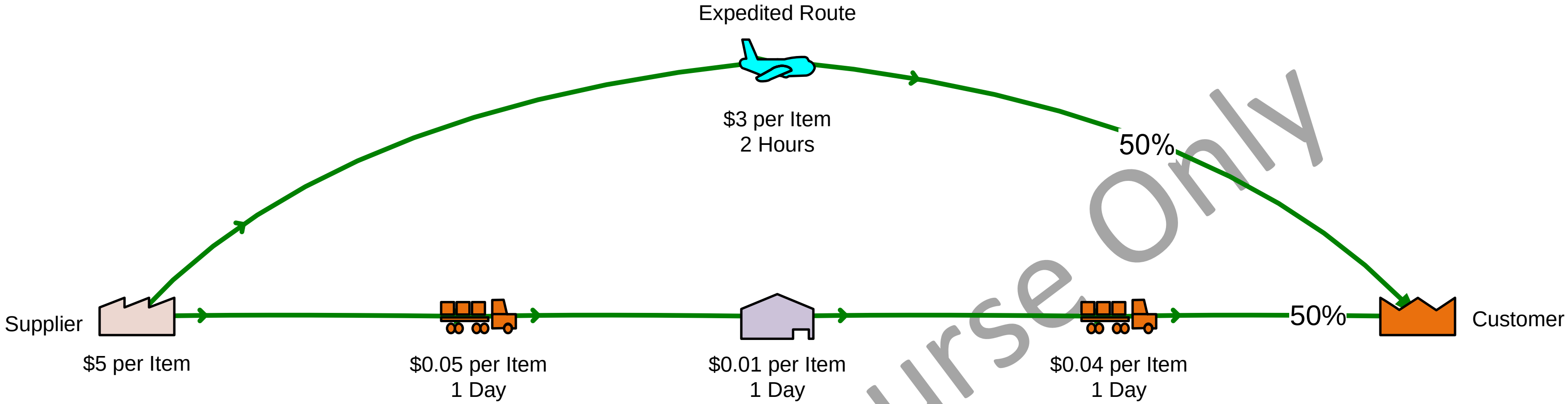
Dual Suppliers



Q. What is the average unit cost of Part A delivered to the customer when demand is sourced from dual suppliers in the percentages shown?

- $2.7 per Item
- $2.4 per Item
- $2.5 per Item
- $3 per Item

Alternate Transport



Q. What is the average cost per part delivered to the customer?

- $13.10
- $5.10
- $6.55
- $8.00

Time and Inventory Quantity Variations

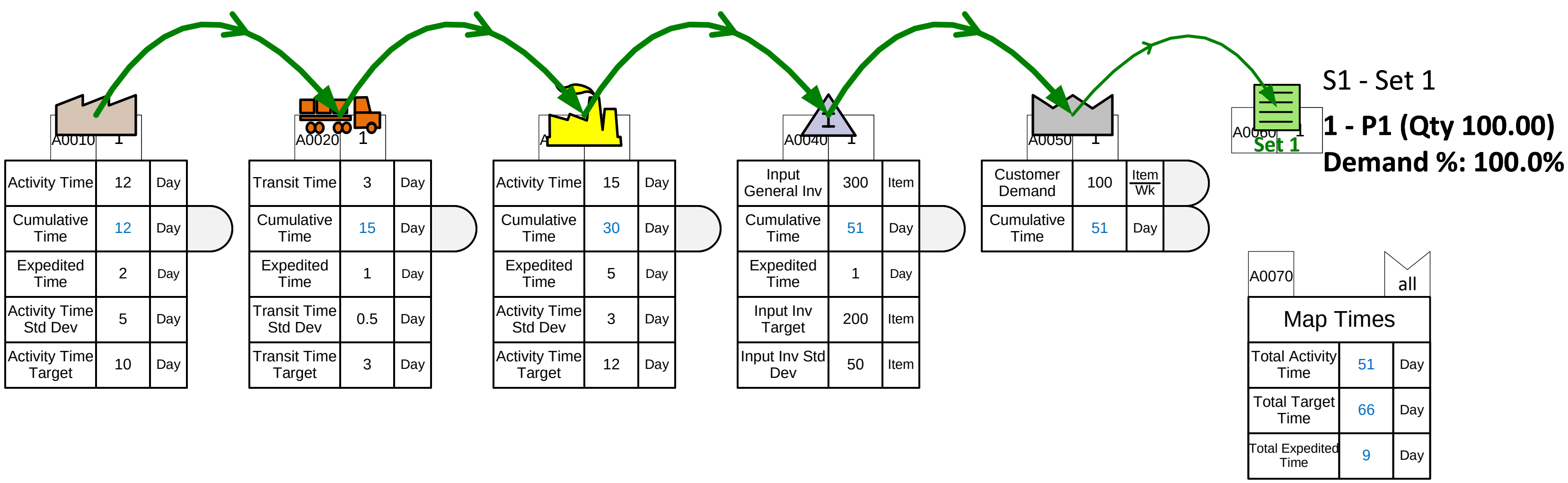
Many of the problems in value streams come down to variation in key parameters. Understanding the variation can help identify what improvements might be possible.

Mix Supply Network has input variables like Activity Time, Transport Time and Inventory Qty. These are used for lead time calculations and charting

The application also allows input of variations in the above values so that they are visible on the map. Variants that can be input and displayed are

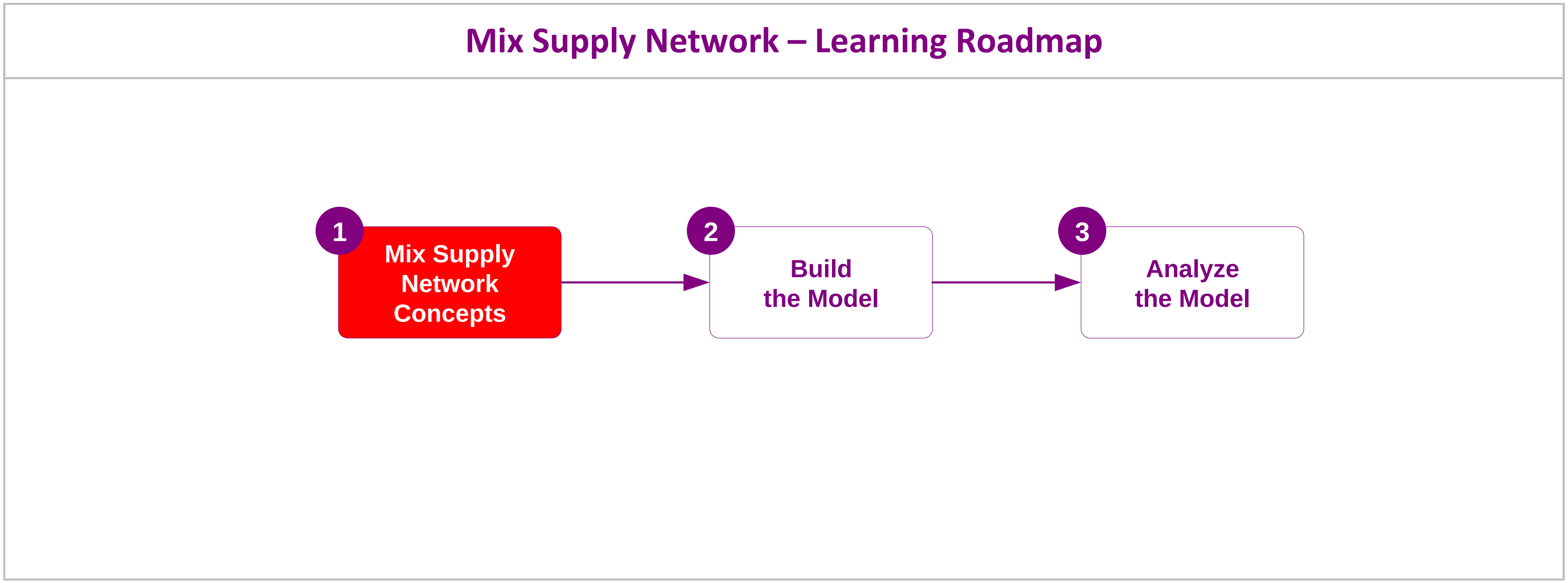
- Standard Deviation
- Target Time
- Expedited Time

Target time and Expedited Time have associated summation calculation across the map such that the sums can be compared to the sum of the current map times. This is illustrated in the example below



You learned:

- Supply network terminology such as activity time, inventory, scrap, activity cost, holding cost, and risk
- How demand flows upstream from the customer
- Cost calculations such as unit cost, value of inventory, and holding cost



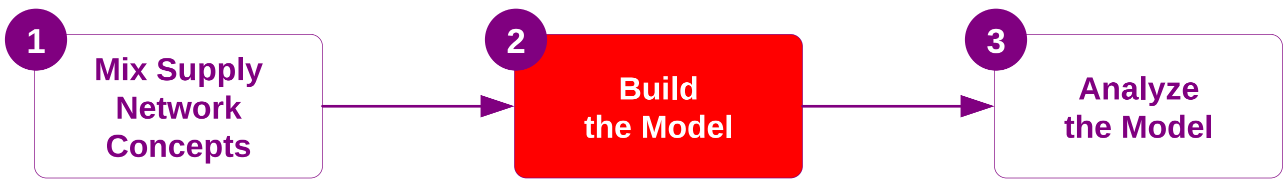
What's next:

Next, you will learn how to declare the products for your value stream, and how to organize them into route sets.

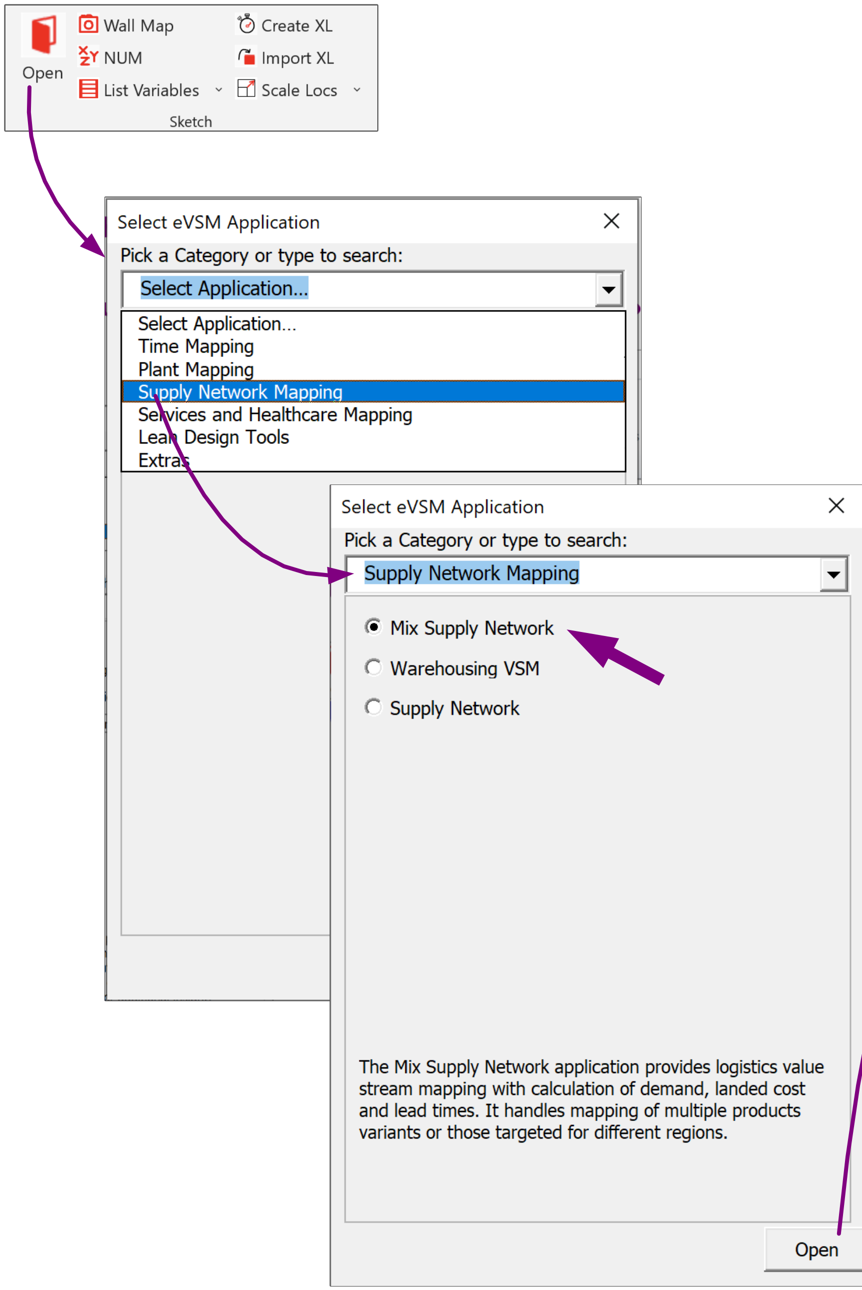
Building a Value Stream Model with Mix Supply Network

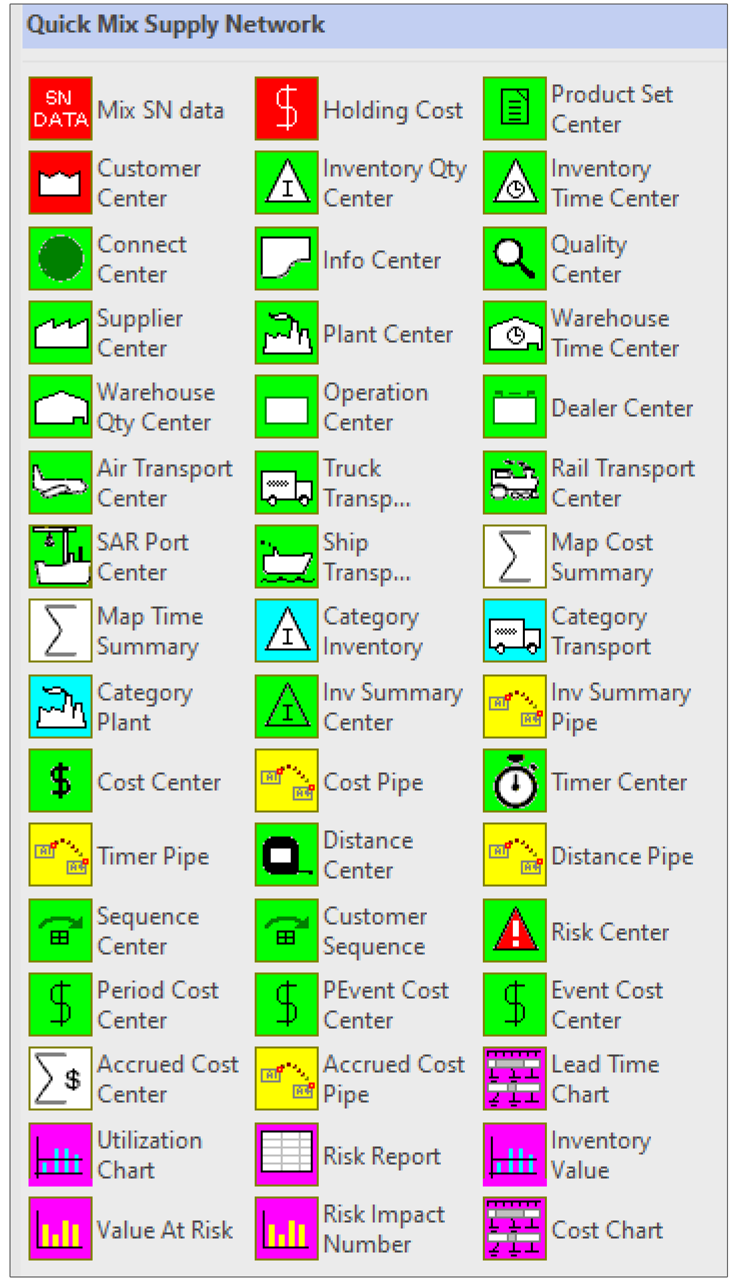
This lesson teaches how to build an eVSM Mix model with the Mix Supply Network application from scratch. A brief overview is followed by descriptions and hands-on exercises to declare a mix of products, draw the flow, specify routings, enter data, check, and solve the model.

Mix Supply Network VSM Modeling Process



Mix Supply Network Stencil



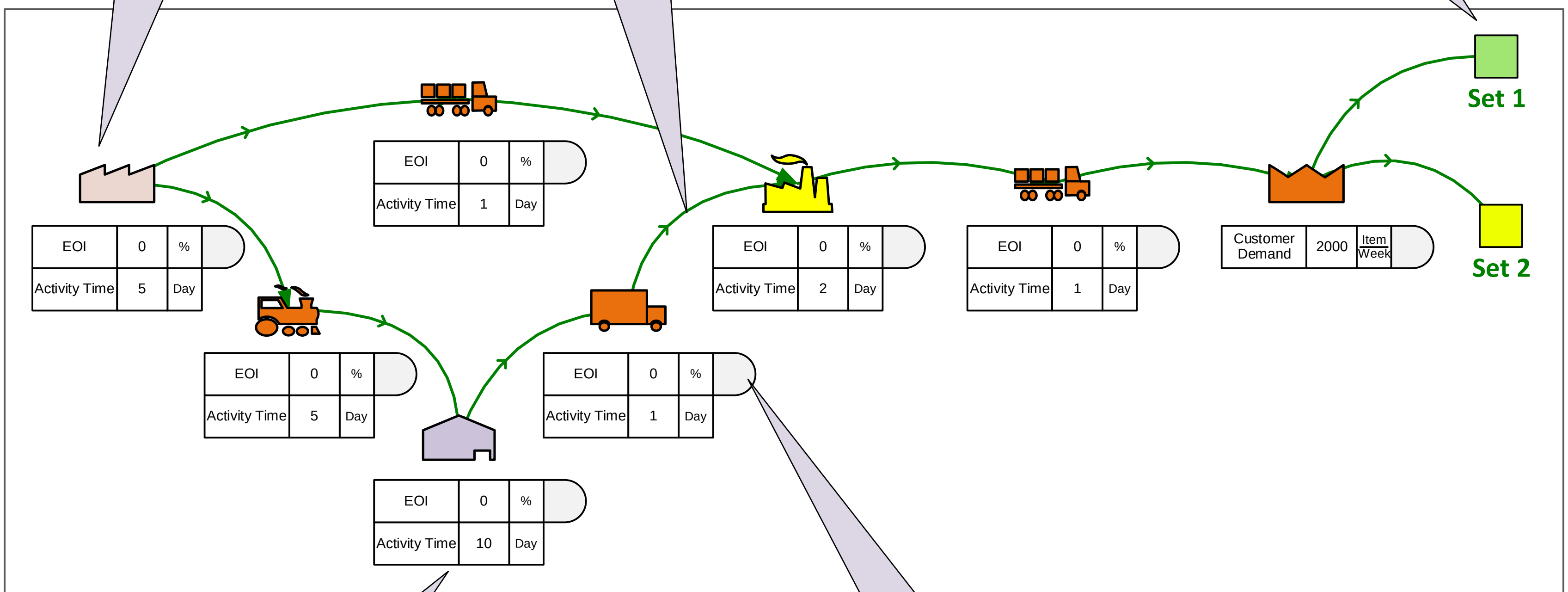


Essential Terminology

Centers
These are the colored icons that represent each node in the network such as plants, inventories, suppliers, customers, transport, etc.

Sequence Arrows
Represent the material flow between centers

Routing Sets
Each routing set represent a group of products which all go through exactly the same nodes



Data Shapes
Have a name, value, and unit to represent data

Product Distribution Tabs
Double-click to view/edit product specific values

Start Mix Supply Network Map

Initiate this page for a Mix Supply Network map

Declare Products and Sets with the Mix Manager

1. Initiate this page for a Mix Supply Network map.
2. Use the Mix Manager dialog (not Excel) to enter the following products and sets.
3. Exit the Mix Manager dialog and click the “Draw Sets” button in the toolbar.
4. Click Grade It.

Set Names	Product Names
China	Product 1 Product 2
USA	Product 3

Edit Products and Sets

In the Mix Manager dialog, to make the following changes. After Editing, click Draw Sets in the toolbar, and then click Grade It in the eLeanor control panel.

Current:

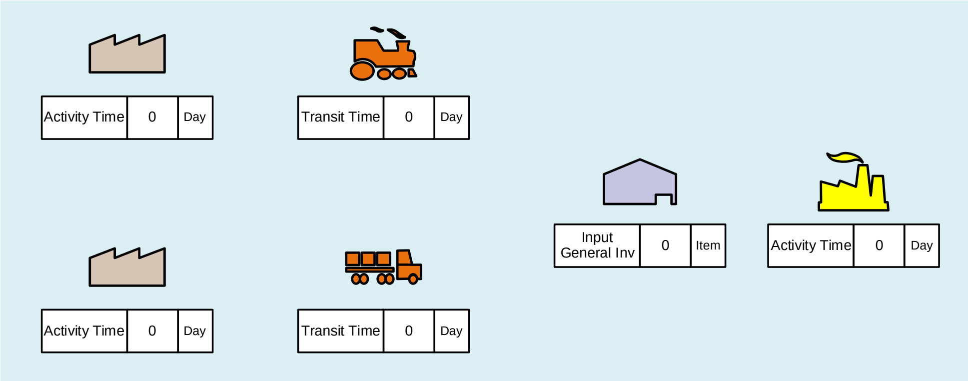
Set Names	Product Names
Set 1	Product 1 Product 4 Product 5
Set 2	Product 2 Product 3

Change to:

Set Names	Product Names
Set 1	Product 1
Set 2	Product 5 Product 4
Set 3	Product 2 Product 3

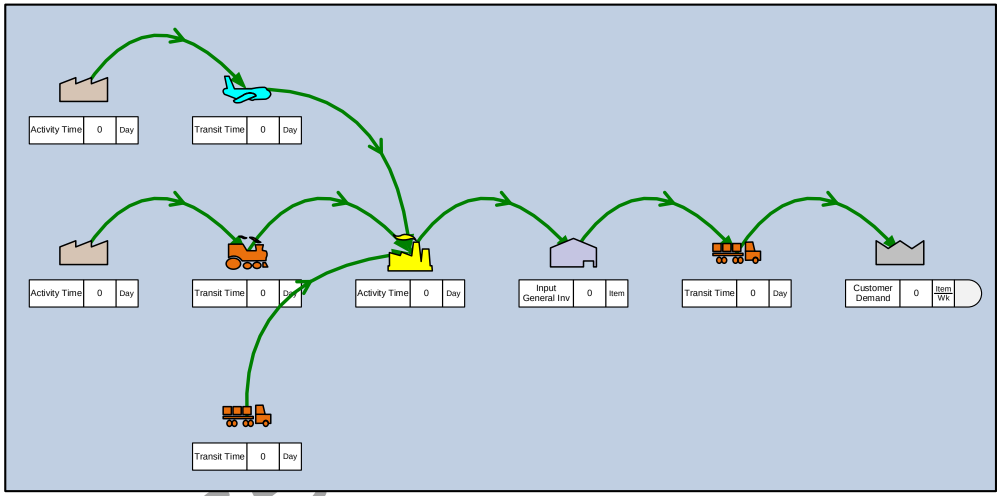
Drop and Arrange the Centers

Find the icons shown in the blue image in the **Quick Mix Supply Network** stencil and drop them on this drawing page



Draw and Sequence

1. Initiate the page for a Mix Supply Network Map
2. Draw this Flow. No need to draw any Set centers.
3. Add Sequence arrows to show the flow of materials.
4. Grade It!

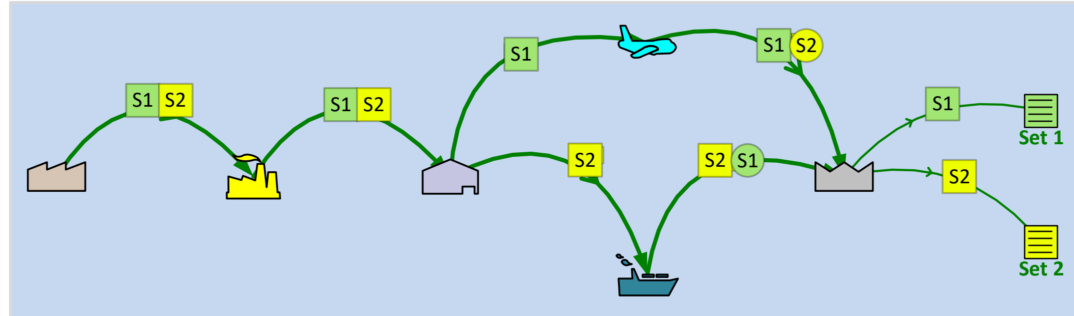


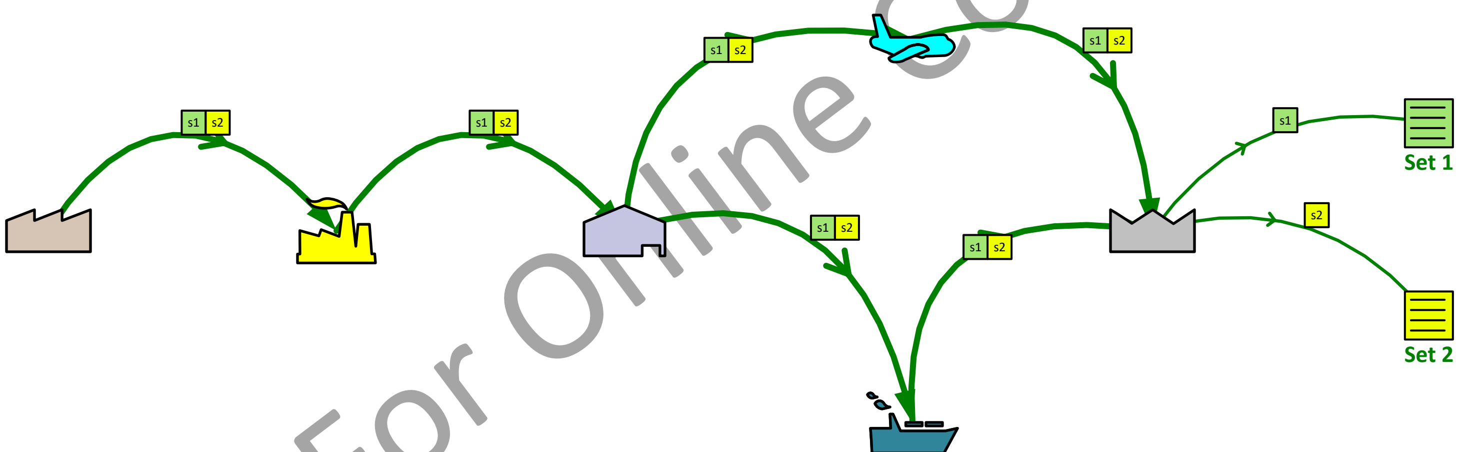
Adjust the Set gates

Use the “Display Gates” button to make the set gates visible on the sequence arrows. Then adjust gates as shown in the image.

Adjusting gates was covered in the Mix Time course. You can download a PDF copy of the Mix Time course notes at

https://evsm.com/MixTimeV12





s1 - Set 1

p1 - Product 1 (Qty 0) Demand %: 0%

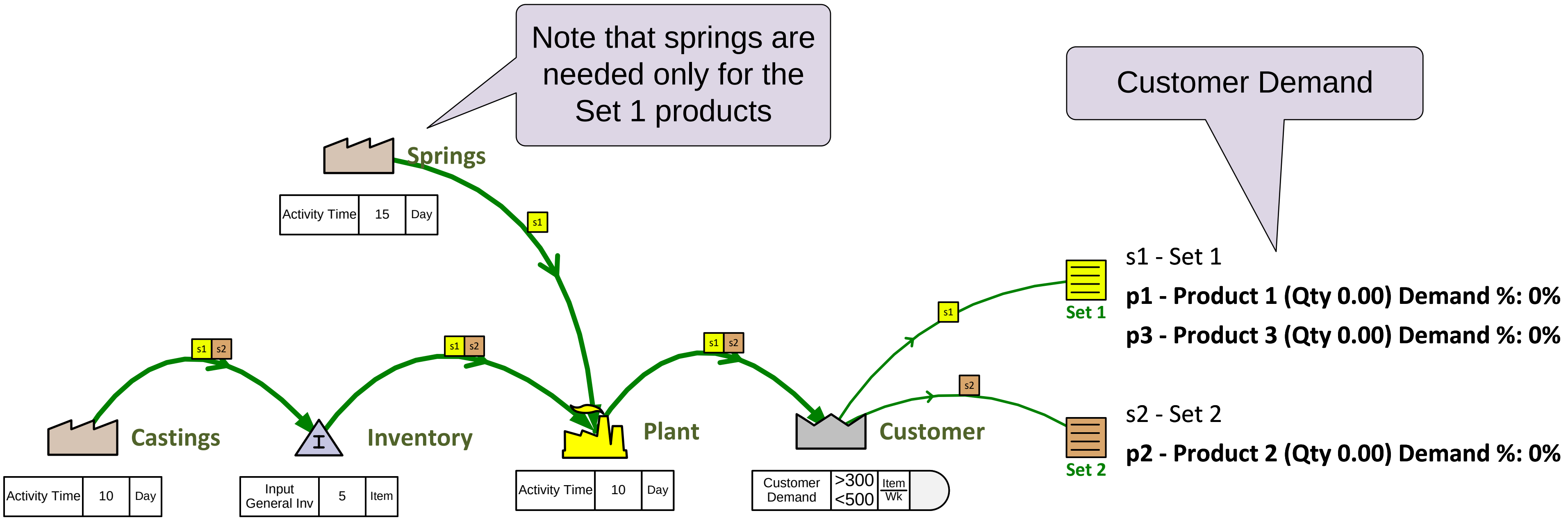
p2 - Product 2 (Qty 0) Demand %: 0%

s2 - Set 2

p3 - Product 3 (Qty 0) Demand %: 0%

Example Map 1

In the next exercise, you will draw the map below from scratch.



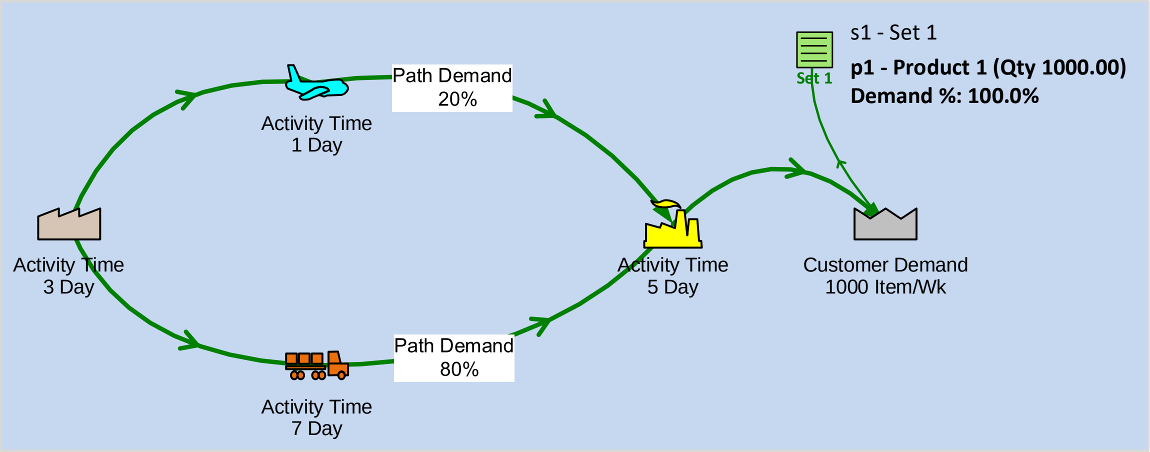
Example Map 1

Follow these steps. If unclear, review lesson 3 in the Mix Time course notes at: https://evsm.com/MixTimeV12

1. Initialize the page for a Mix Supply Network Map
2. Enter the products and sets in the Mix Manager dialog
3. Draw the flow including the sequence arrows
4. Specify the routings with the set gates
5. Enter the data values

For your convenience, an image of the completed map is available above this page.

Example Map 2



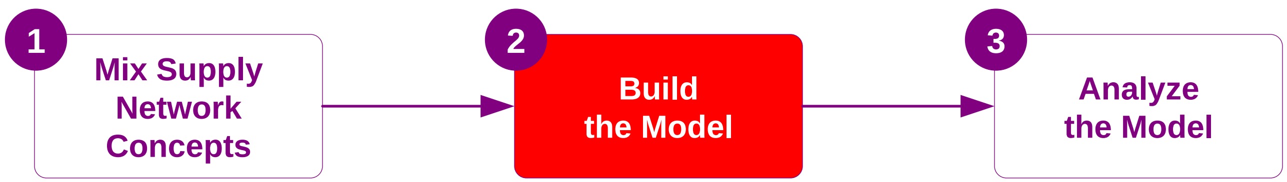
1. Initialize the page for a Mix Supply Network map.
2. Declare the single product and set through the Mix Manager.
3. Draw the flow and connect with sequence arrows.
4. Enter the data values.
5. Make the "Path Demand %" variable visible on the sequence arrows and enter the 20% and 80% values.
6. Make the "Lead Time" variable visible on the Customer center

6. Solve the model, verify the Lead Time is 15 Days, and Grade It!

You learned:

- How to declare products directly in Visio and through Excel
- How to draw and sequence the flow of materials
- How to specify routings with the set gates
- How to check and solve the model

Mix Supply Network VSM Modeling Process



What's next:

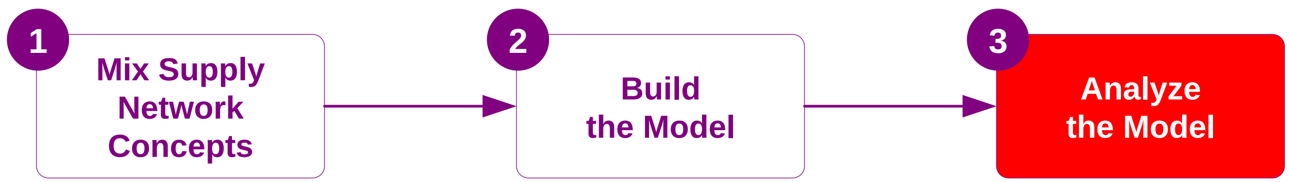
You will learn how to visualize the model results with charts and gadgets, and how to explore "What-If" scenarios with supply network models.

Analyzing Mix Supply Network VSM

In the last lesson you learnt how to build a working value stream model with the Quick Mix Supply Network application.

In this lesson, you will learn how to adapt and analyze the model for key scenarios and metrics such as demand, lead time, capacity, inventory and cost.

Mix Supply Network – Learning Roadmap

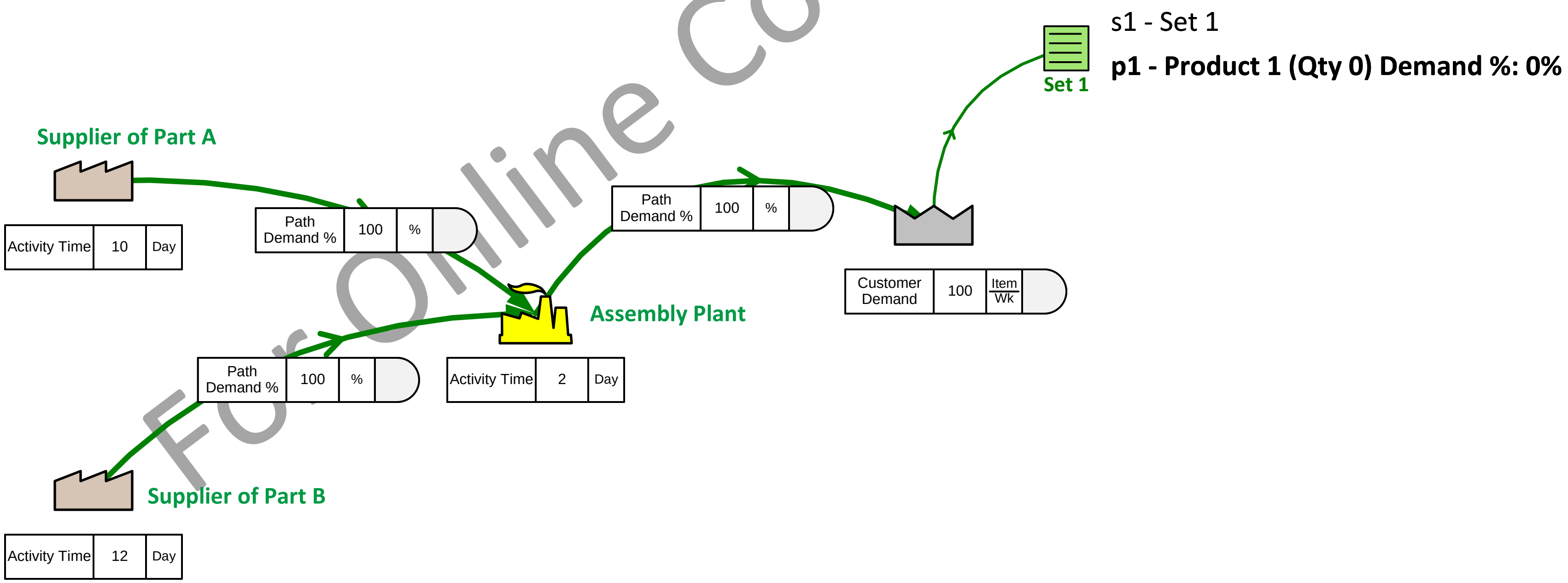


Establishing Component Demand Relative to Finished Goods

The final assembled product needs 2 of part “A” and 3 of part “B”. Based on the current customer demand, we wish to use the VSM to show the demand for the parts

Follow these steps:

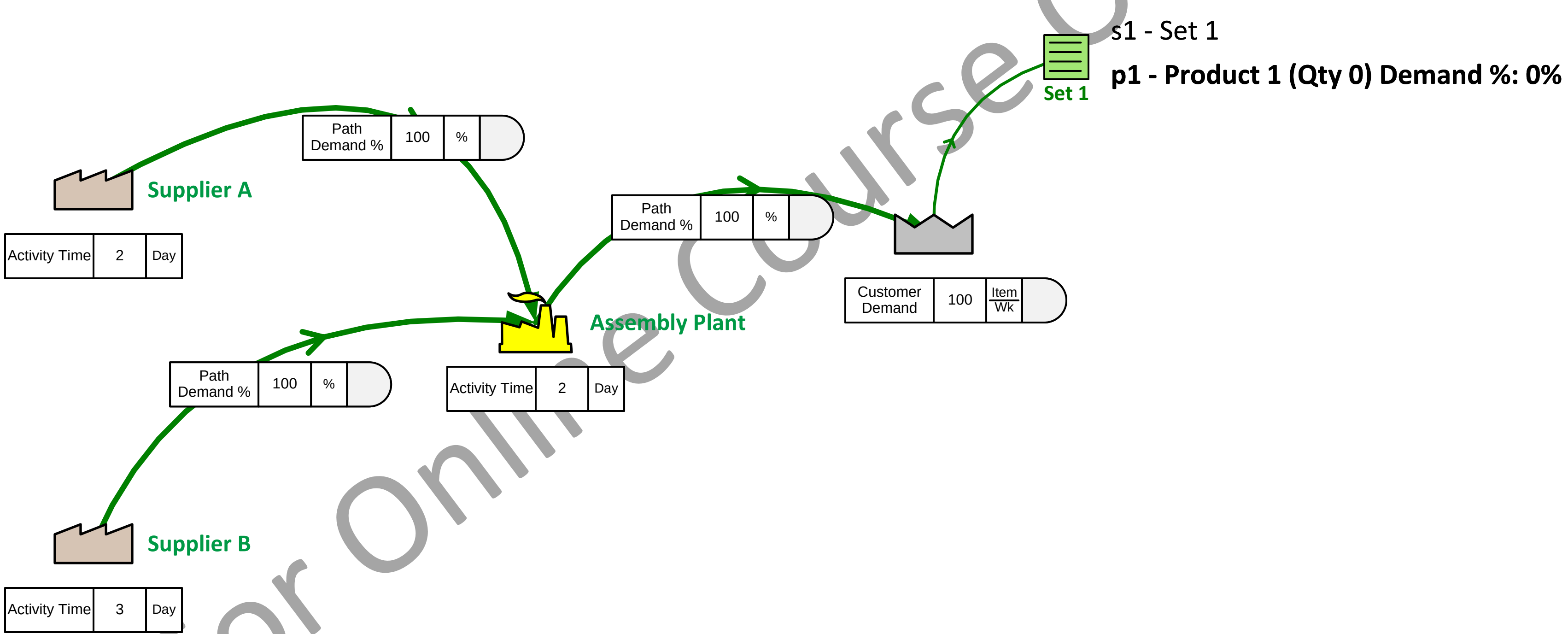
1. Use the “Path Demand %” values to establish upstream demand versus downstream demand.
2. Through the Views button unhide the “Sum Activity Demand” variable.
3. Solve the Map to show the demand for the parts.



Impact on Upstream Demand of Downstream Scrap

ModIfy the map below to show there is 10% scrap at the Assembly plant.

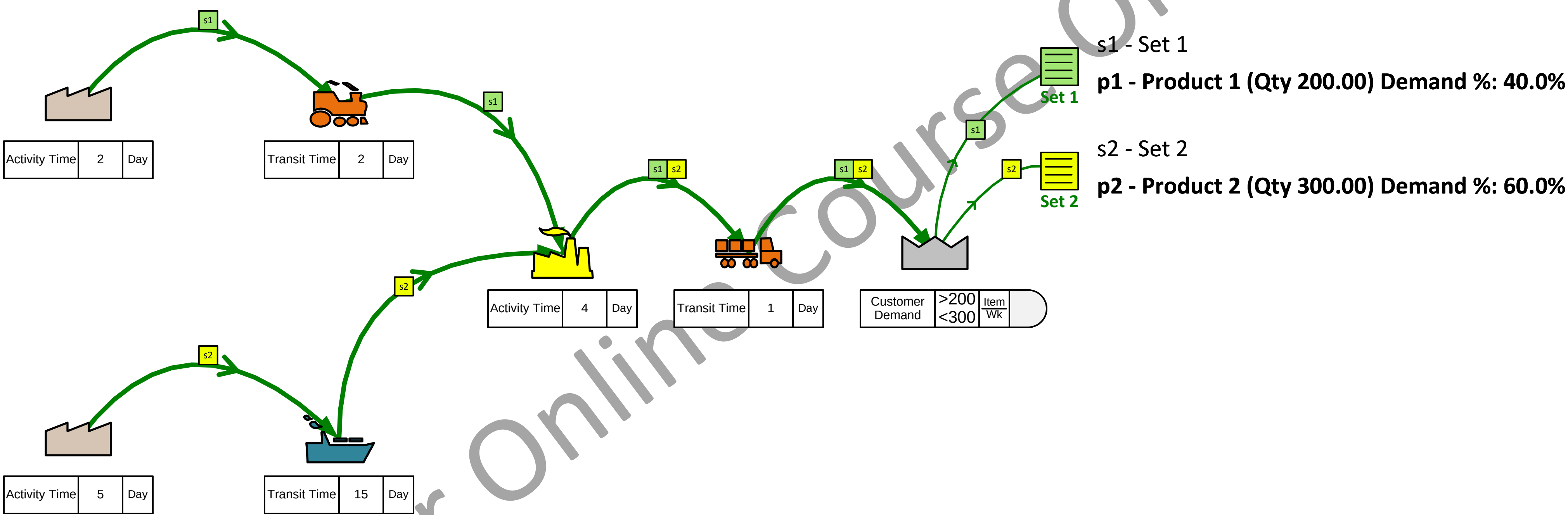
1. Make the “Scrap” and “Sum Activity Demand” variables visible.
2. Modify the “Path Demand %” to show that each final product unit requires 4 parts from supplier A and 3 parts from supplier B.
3. Solve and check the supplier demand for the parts.



Plotting a Lead Time Chart for A Set

Follow these steps:

1. Drag out the lead time chart control shape from the Quick Mix Supply Network Stencil.
2. Right click on path filter (top right of the chart shape) to “Add Connector” and then glue this to the Set 2 shape
3. Click the Autopath button in the eVSM toolbar to set the path numbers for the chart control (to those corresponding to Set 2)
4. Right click the chart control and select “Plot Ladder Chart”

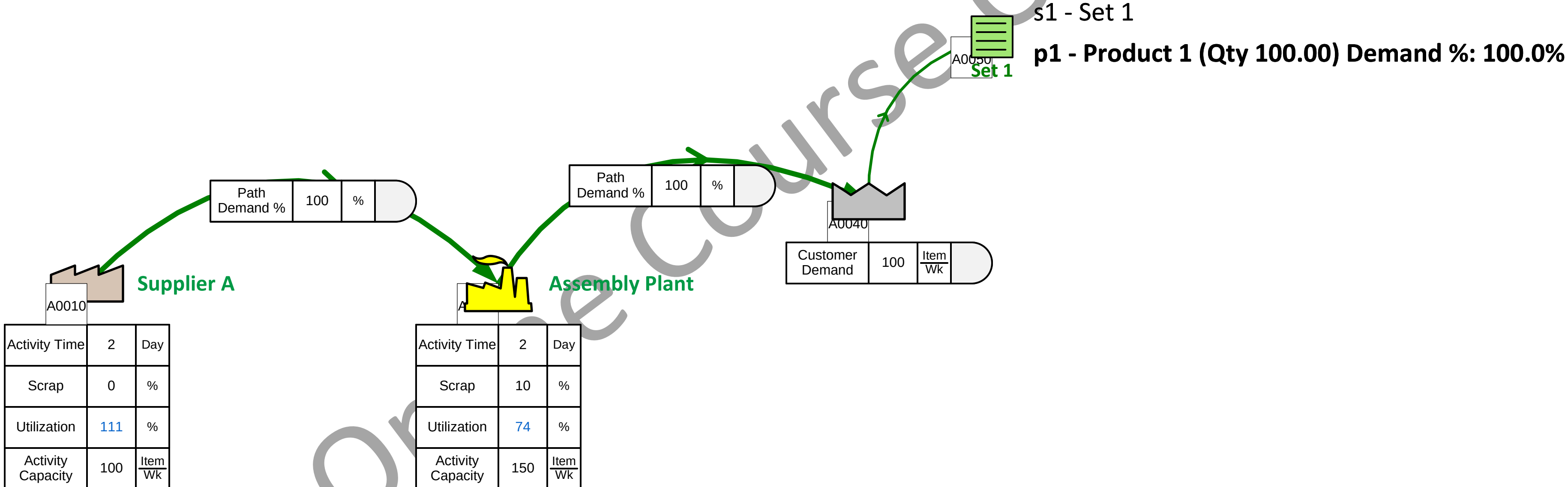


Managing Capacity – Alternate Suppliers

There is 10% scrap at the plant and Supplier A not able to fulfil the demand. Our short term plan is to add a second supplier (Supplier B) who will handle 20% of the demand

Follow these steps:

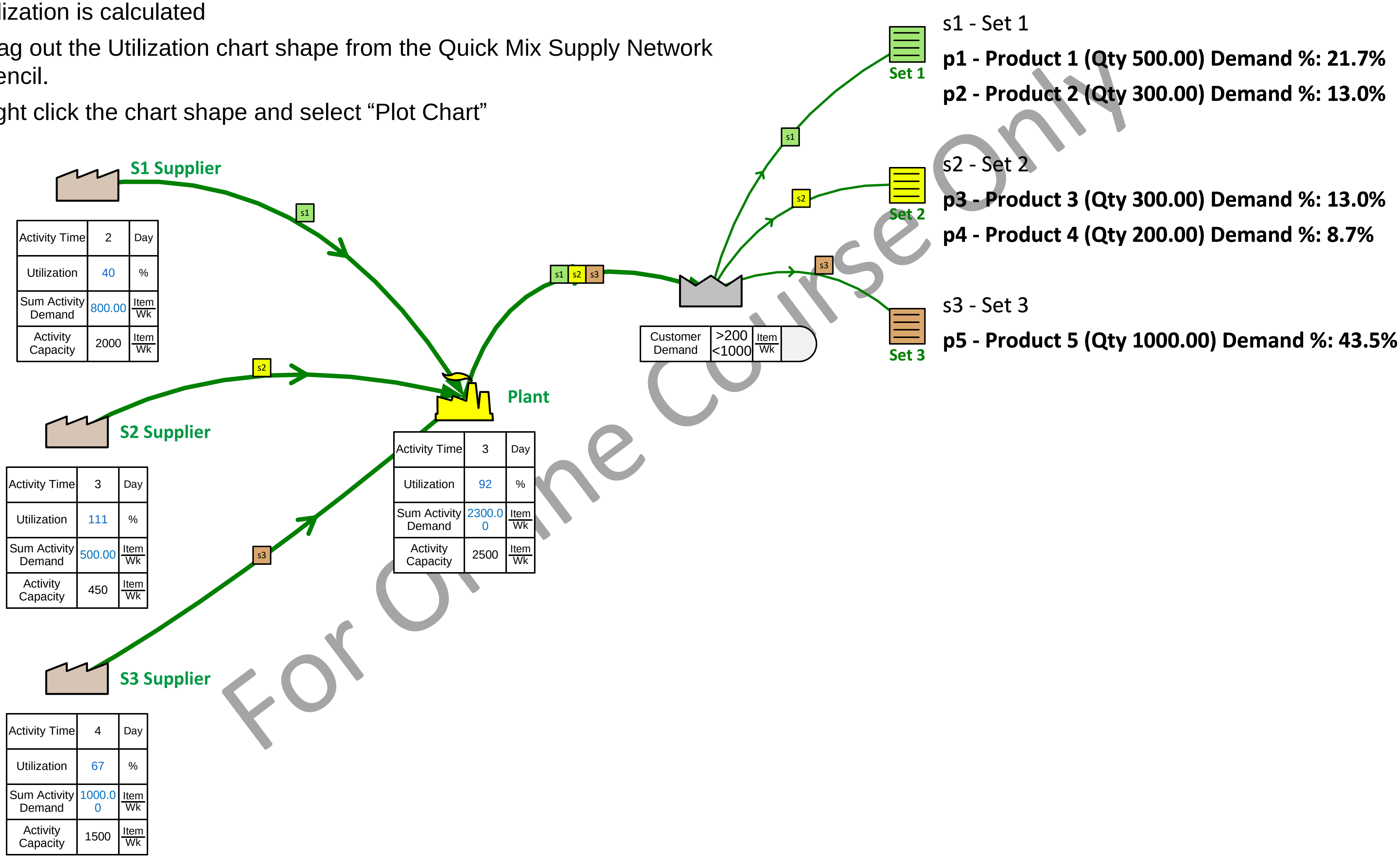
1. Add the second supplier
2. Modify the path demand % values
3. Solve the map and check the new supplier utilizations



Plotting a Capacity Utilization Chart

Follow these steps:

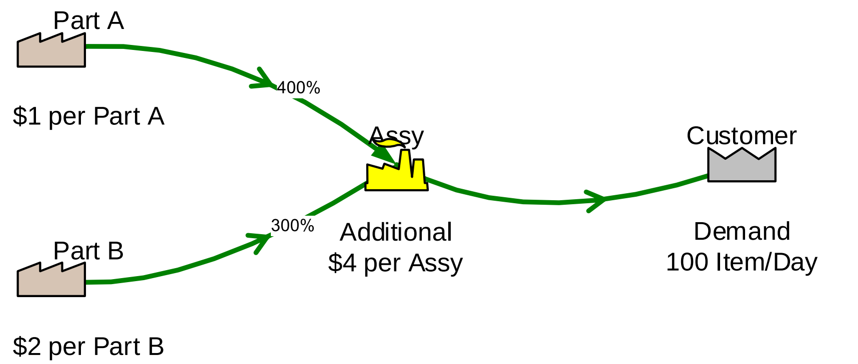
1. Compare the demand and capacity numbers at nodes to understand how utilization is calculated
2. Drag out the Utilization chart shape from the Quick Mix Supply Network Stencil.
3. Right click the chart shape and select “Plot Chart”

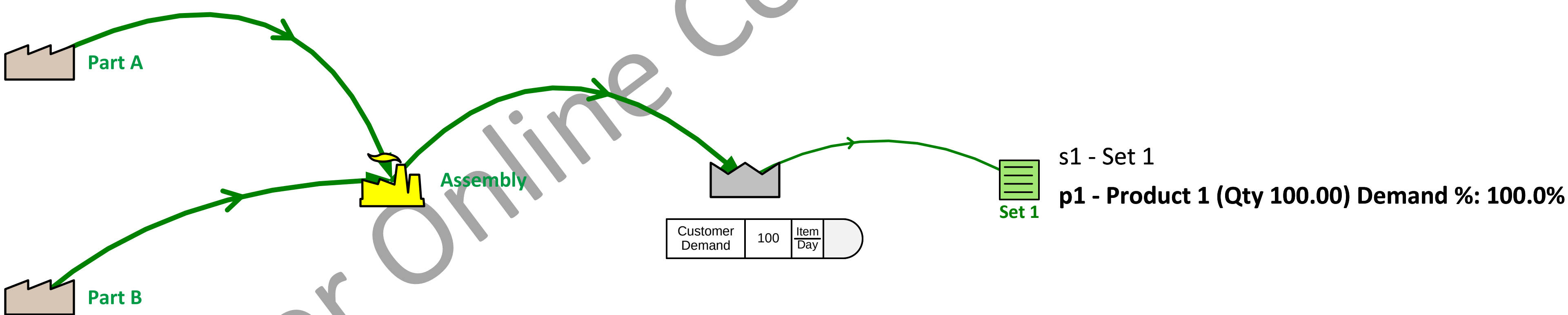


Add Cost Data and Creating a Cost Summary for Map

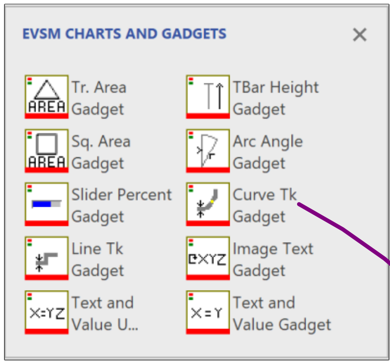
Follow these steps:

1. Make the “Added Unit Cost” and the “Path Demand %” variables visible.
2. Enter the data shown in the image.
3. Drag out the “Map Cost Summary” from the Quick Mix Supply Network stencil. Set the unit for the “Total Activity Cost” variable to $/Day
4. Now use the “Solve” button to calculate the total activity cost per day





Line Thickness Gadget to Visualize the Cash Flow

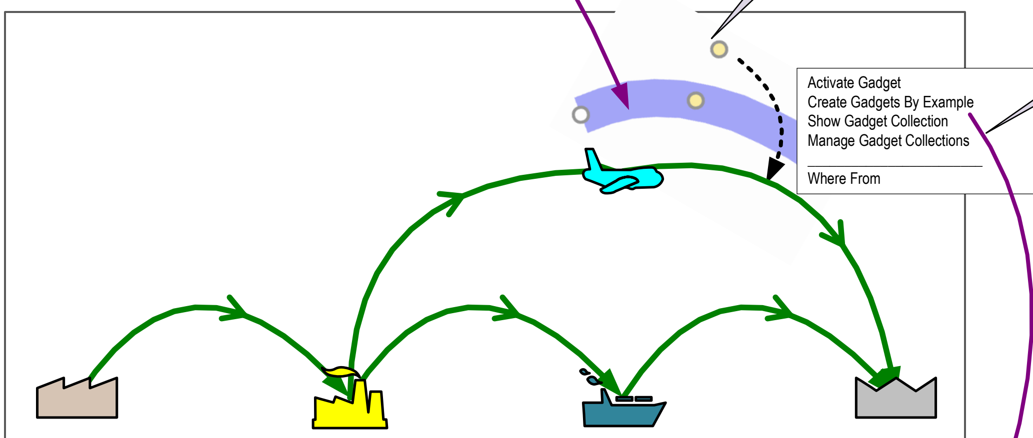


1 Drop a Curve Thickness gadget anywhere on the page

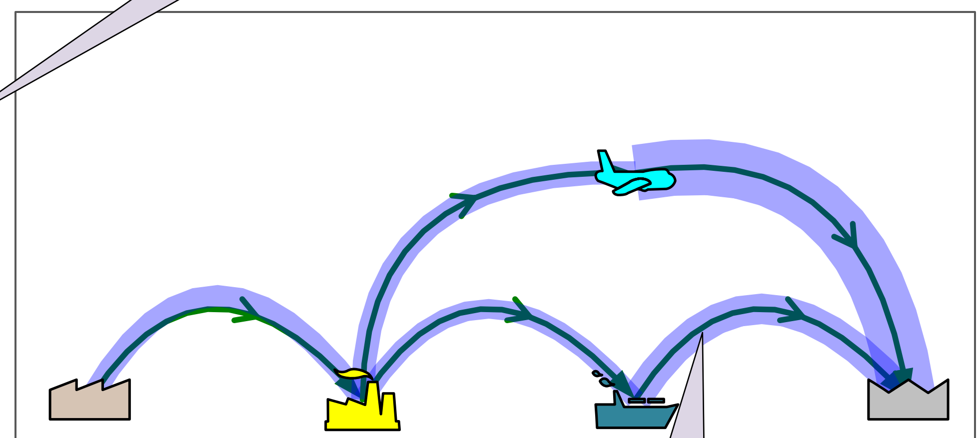
2 Glue it’s floating glue-point to any one sequence arrow

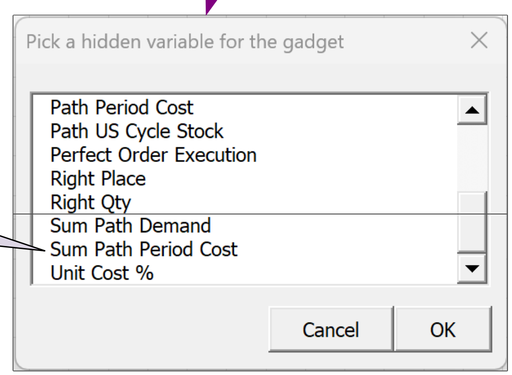
3 In the right-mouse-menus, click the “Create Gadget By Example” command

Before



After



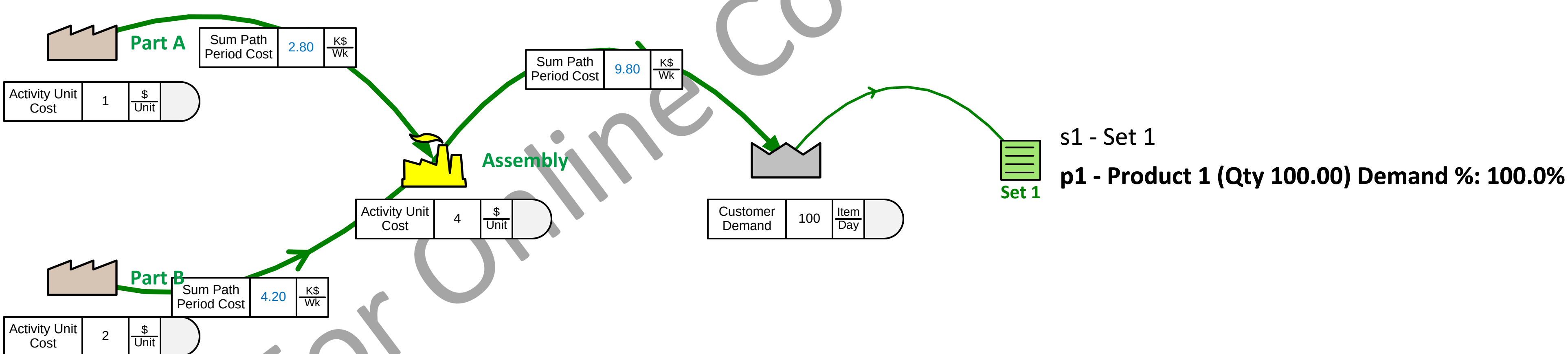


4 Select the “Sum Path Period Cost” variable and click OK

The gadget thickness represents the total demand flowing through each sequence arrow

Visualize the Cash River for this Supply Network

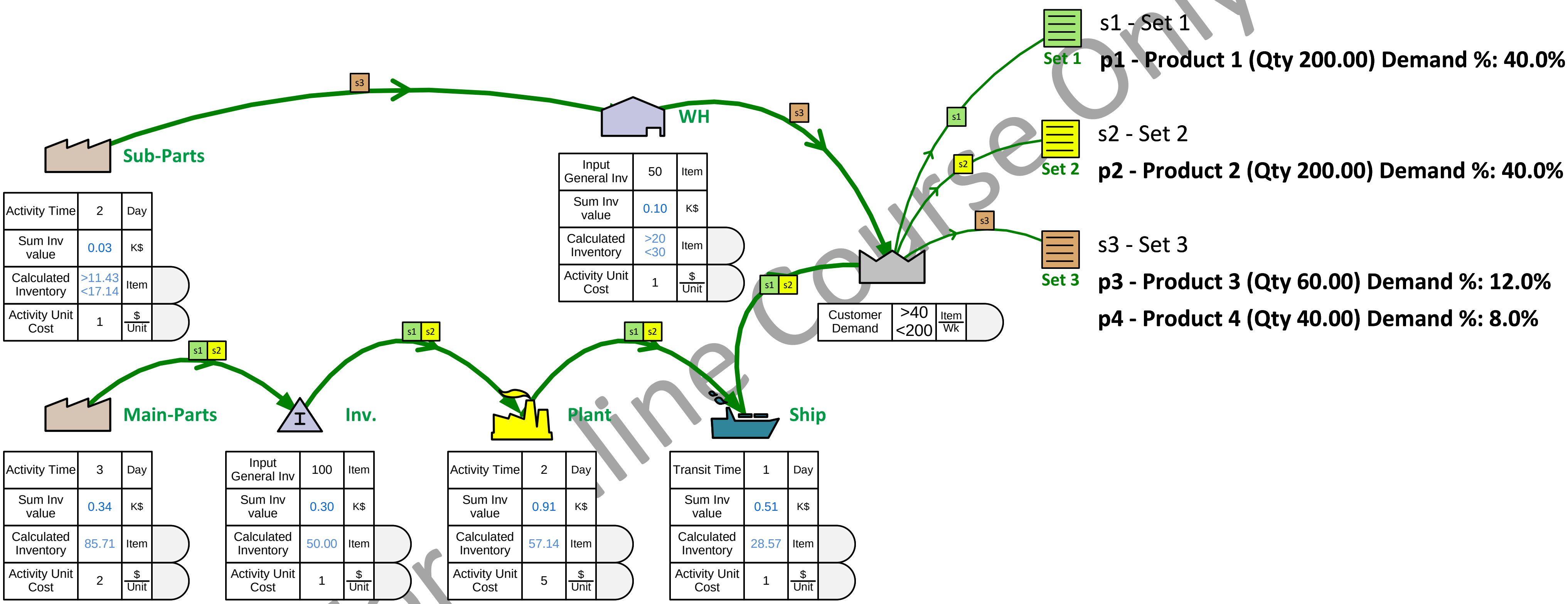
Follow the steps shown on the previous page to visualize the "Sum Path Period Cost" with the Line Thickness gadget.



Plot the Inventory Value Chart

Follow these steps:

1. Look at the Calculated Inventory and Sum Inv Value numbers and understand how they are computed
2. Drag out the Inventory Value control shape from the Quick Mix Supply Network Stencil.
3. Right click the chart control and select “plot..”

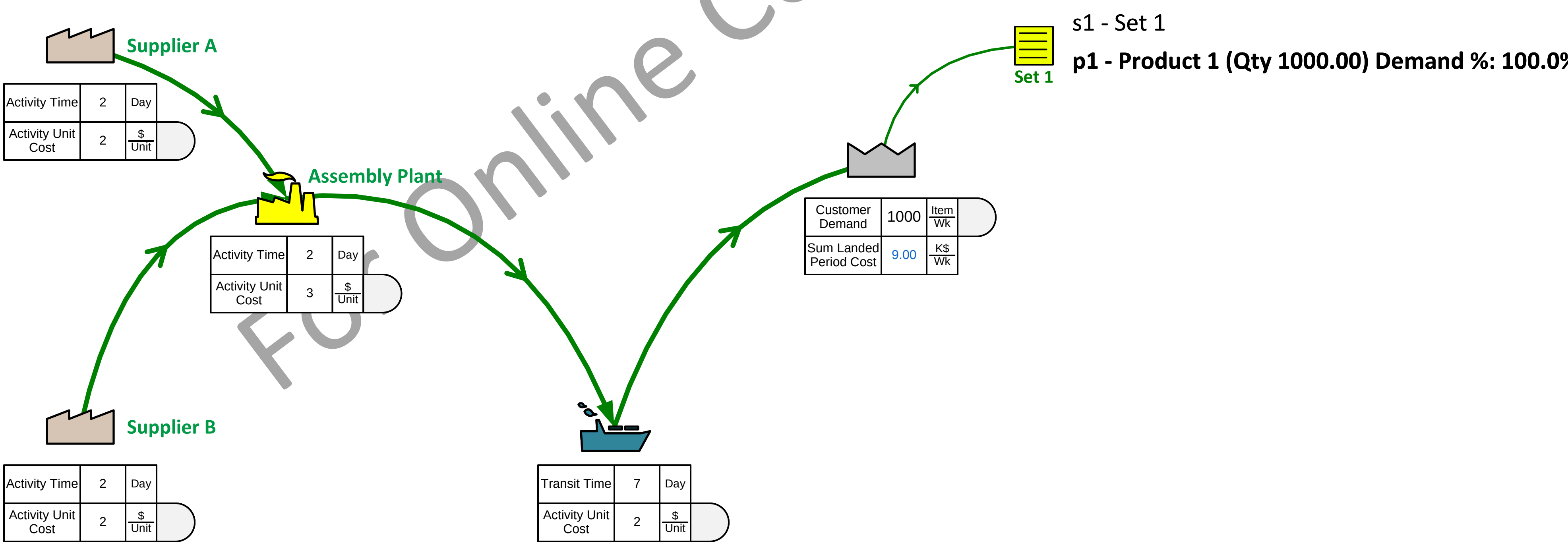


Managing Expedited Transport and Costs

The customer wants a second, expedited transport route, via air. The transit time = 1 day, cost = $5/Unit. They estimate around 25% of the demand via this route. Modify the model to check how much the weekly “Sum Landed Period Cost” will go up by.

Steps:

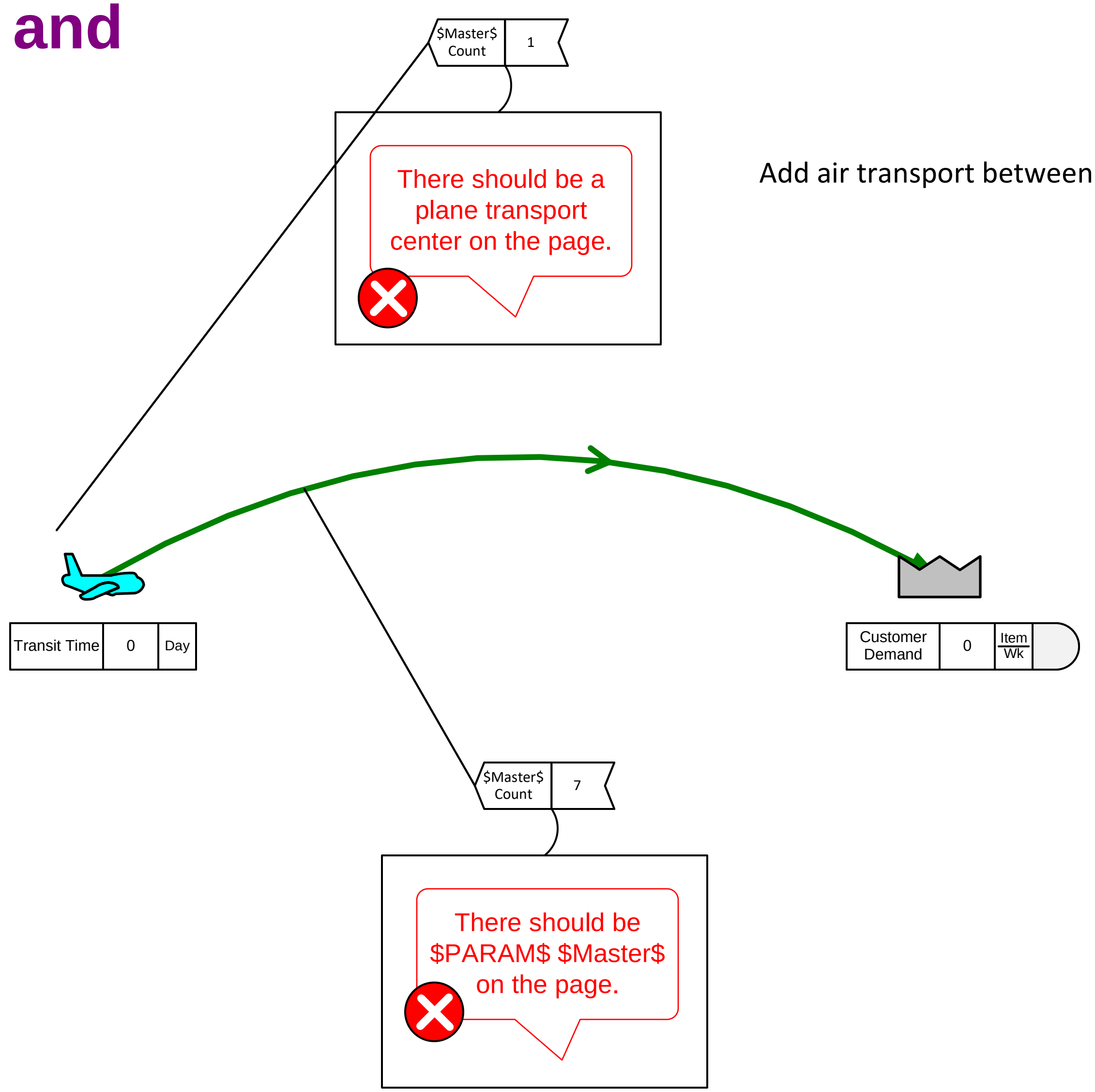
1. Add an Air Transport center with the above data.
2. Make “Path Demand %” variable visible on the sequence arrows and enter the 25%/75% demand split.
3. Solve the model and check the “Sum Landed Period Cost” at the Customer center.



Q Pt’s	10

Manage Transport and Associated cost

Add air transport between plant and customer

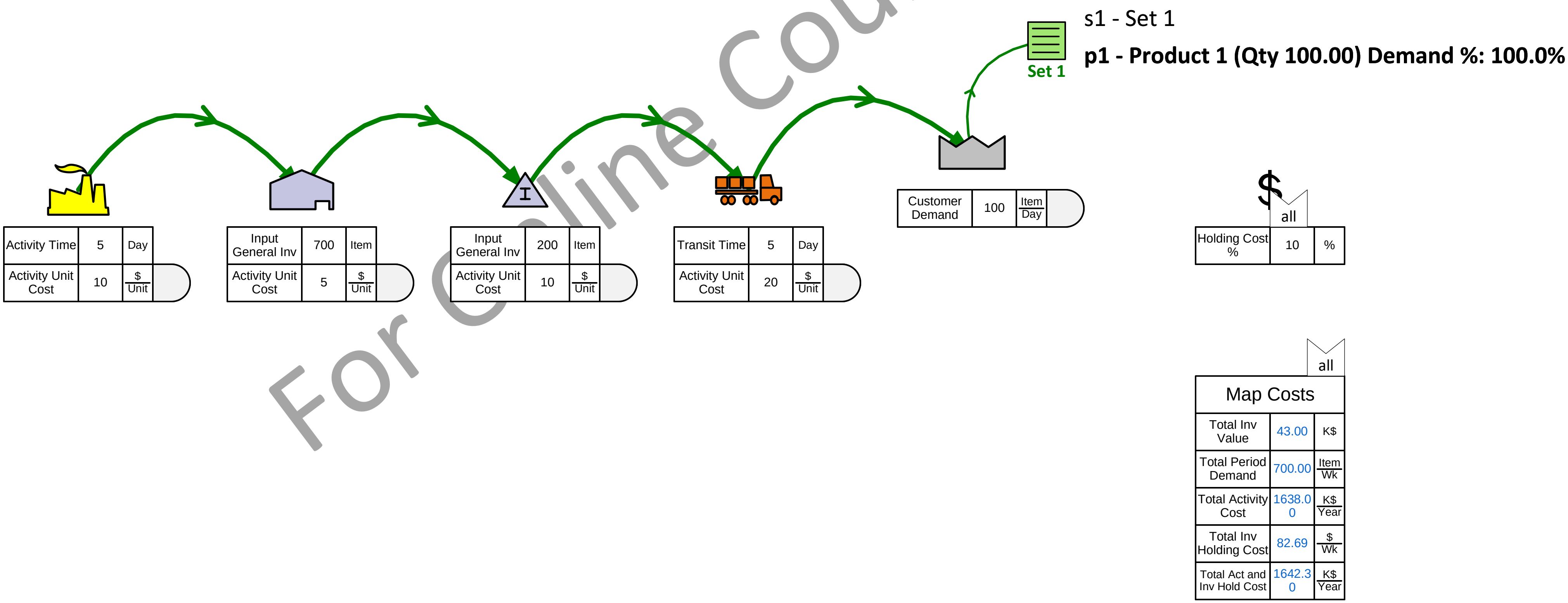


Text Block

What is the Inventory Holding cost per week?

If the transport is delayed by 2 days what will be the inventory holding cost per week ?

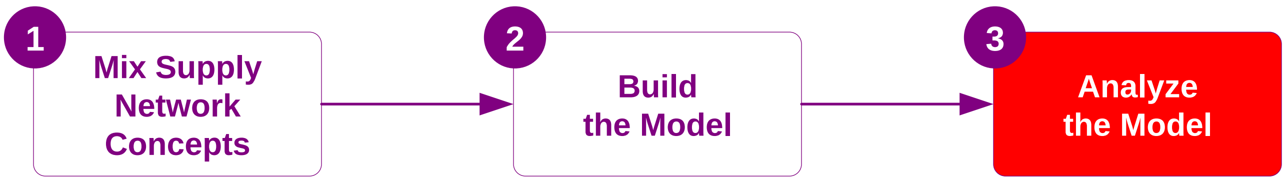
- 100 $/week
- 150 $/week
- 85 $/week
- 90 $/week



You learned:

- How to adapt and analyze the supply network model for key scenarios and metrics such as demand, lead time, capacity, inventory and cost.

Mix Supply Network – Learning Roadmap



What's next:

Attempt your first map with Mix Supply Network VSM and contact support@evsm.com with any questions about this course and about your value streams.

Useful Links

eVSM Toolbar Guide	evsm.com/toolbarguide
eVSM Productivity Guide	evsm.com/productivity
eVSM Blogs	evsm.com/blog
eVSM Support FAQ	evsm.com/support
Download the Latest Version	evsm.com/install